

Residue fluid catalytic cracking: A review on the mitigation strategies of metal poisoning of RFCC catalyst using metal passivators

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Abstract

The workhorse residue fluid catalytic cracking catalyst used to process heavy feedstocks, such as atmospheric residue, has good catalytic activity, selectivity, hydrothermal stability, fluidability, and coke selectivity. Processing these heavy feedstocks has several challenges due to heteroatoms such as vanadium, nickel, sodium, iron, sulfur, and nitrogen. To give perspective on these challenges, motivation for and recent technologies for Residue Fluid Catalytic Cracking (RFCC) have been reviewed. Also highlighted are the relevance of (i) multi-zone coordinated-controlled, (ii) two-stage riser (conventional), (iii) two-stage riser (for propylene production), and (iv) auxiliary riser (gasoline modification) on reactor performances. These heteroatoms are detrimental to the catalysts as they act to either permanently or temporarily deactivate them, thereby promoting undesirable reactions such as dehydrogenation and coke production. Furthermore, the impact of the catalytic cracking catalyst selectivities on low molecular weight production, especially propylene and other olefins, has been pointed out. It is, therefore, important to develop a catalyst with improved metal tolerance while maintaining its intrinsic properties. Antimony, bismuth, boron, magnesium, tin, and rare earth compound (cerium, lanthanum) have attracted attention recently due to their abilities to trap and passivate Ni, V, Na, and Fe against their harmful effects on the catalyst.

Keywords:

Residue fluid catalytic cracking, RFCC Catalyst, Deactivation, Vanadium, Nickel, Passivators

29 1 Introduction

30 Fluid catalytic cracking (FCC) is a secondary conversion process in the petroleum
31 refining industry used to convert high-molecular-weight, high-boiling fractions of petroleum
32 to valuable fuels such as gasoline, diesel, olefinic gases, and liquefied petroleum gas (LPG).
33 Globally, 59 countries have active refineries with FCC and a total refinery FCC unit capacity
34 of 14,259 mbd in 2020 [1]. The biggest refineries in the world in terms of the refinery FCC
35 unit capacity in 2020 include Baton Rouge (US), Baytown (US), Jamnagar I (India),
36 Jamnagar II (India), and Texas City III (US) [1]. Some of the largest upcoming refineries
37 with FCC units in the world during the outlook period of 2021-2025 include the Dangote
38 refinery, Lagos (Nigeria), Yulong (China), and Turban II (Indonesia) [2].

39 The FCC unit is regarded as the most important conversion unit and the cash cow of
40 the refinery because the fuel produced from this unit alone accounts for 35 to 50% of the total
41 gasoline produced globally in the petroleum industry [3–6]. However, the world crude oil
42 supply changes towards heavier crudes, coupled with the quest by refiners to process cheaper
43 crude to increase refinery margins, and the increasing demand for propylene and
44 transportation fuel, have led to the development of the residue fluid catalytic (RFCC) process
45 to convert residue feedstocks. A shared setback established in both the FCC and RFCC units
46 is the regular decrease of catalytic performance caused by the gradual deposition of
47 contaminant metals such as Ni, V, Fe, and Na on the catalyst during the processing of heavy
48 hydrocarbon feedstocks [7–12]. These metals destroy the catalyst and promote undesirable
49 side reactions, such as producing large amounts of coke and hydrogen. A large quantity of
50 hydrogen is undesirable because it affects downstream equipment such as wet gas
51 compressors (WGC). The WGC is volume-limited and not designed to handle gases beyond a
52 certain volume. Production of a large amount of coke upsets the reactor and regenerator heat

balance as well as affects the main air blower (MAB), which like WGC, is also limited in the amount of air it takes to burn off the coke in the regenerator during the regeneration process and leads to a reduction in feed throughput. In addition to upsetting the heat balance and the MAB, a large amount of coke temporarily deactivates the RFCC catalyst by blocking the active sites, thereby reducing the catalyst activity. A way to possibly overcome such a setback is by deploying environmentally friendly passivating agents in the RFCC catalyst mix.

Over the years, several authors have published reviews on the conventional FCC unit, including processes, catalyst materials, poisons, and deactivation mechanisms. Vogt and Weckhuysen, [13] reviewed the developments of FCC, the so-called grand old lady of zeolite catalysis. The study discussed the general background of the FCC process, including a short history, process description, chemical reactions, reactor design, and catalyst material. The study also evaluated the means to improve the stability of zeolite structure, propylene selectivity, overall catalyst accessibility, and a state-of-the-art micro-spectroscopy technique for characterising FCC catalysts at the single particle. The findings, however, suggested that shifts in the availability of feedstocks, including tight oils, and renewables, coupled with longer-term demand for gasoline, middle distillates, and propylene, require further developments in both catalysts and process.

Bai et al. [14] reviewed the contamination of fluid catalytic cracking catalysts describing the status of FCC technology, such as catalysts, the variation in feedstocks, and the effects of contaminants of the FCC catalysts. The deposition mechanisms of multiple impurities on cracking catalysts during the FCC process, including nitrogen, sulfur, metals, and coke originating from the feedstock during the FCC reaction, were also discussed. It was reported that vanadium, nickel, iron, and sodium are the metal contaminants having the most harmful effects that can significantly impact the catalyst structure and performance. It was

also found that nitrogen and sulfur are considered less harmful compared to metal contaminants. Pore blockage by the coking effect of sulfur and acid site neutralization by nitrogen are serious problems too.

A review of the deactivation of FCC catalysts by cyclic propylene steaming (CPS) presented some modifications made to the CPS method to further decrease the destructive effects of vanadium on FCC catalysts during deactivation [15]. The methods comprise (i) adjusting the ratio of the time the catalyst spends in reducing and oxidising atmospheres during the CPS cycles and (ii) exposing the metalated catalyst to pre-stabilisation steps with reduction-oxidation cycles. Comparing the activities and selectivity data of FCC catalysts with (a) age distribution and (b) CPS revealed that both have similar catalyst rankings. Moreover, reduction-oxidation cycles were found to simulate the deactivation of FCC catalysts in the absence of contaminant metals better than traditional deactivation in a 100% steam atmosphere.

FCC catalyst deactivation was also reviewed by O'Connor and Pouwels [16] with a focus on the complexity of the inter-relations in catalyst deactivation by (i) ageing, (ii) poisoning and fouling, (iii) the effect on heat balance and catalyst circulation rate, and (iv) the quantification and modelling of FCC catalyst deactivation. The findings suggest that for catalyst poisoning at high metal contents and hydrothermal ageing with low metals, cyclic deactivation of the catalyst in numerous age fractions is the preferred route for a realistic simulation and quantification of the deactivation phenomena.

Patrick-Walker and Peterman [17] reviewed how the contemporary residue FCC (RFCC) units set a new standard for propylene production by exploring the technical challenges and opportunities in targeting high propylene production from RFCC units. The study summarised the advances in RFCC operations and reliability. The differences between RFCC operations and conventional FCC based on the catalysts, feed properties, operating

parameters, and equipment design were also evaluated. The authors pointed out that the processing of heavily contaminated residue using the Universal Oil Products (UOP) combustor type regenerators with catalyst coolers is gaining wider acceptance alongside two-stage regenerators with catalyst coolers due to their ability to withstand the high temperature that results from high Conradson carbon residues present in the residue feedstock.

In an attempt to discuss the catalyst structure suitable for the large molecular weight feed to the RFCC unit, Ishihara [18] reviewed advances in the preparations and reactivity of hierarchical zeolite catalysts for the catalytic cracking of carbonaceous materials. The carbonaceous materials include simple straight-chained and substituted-aromatic hydrocarbons, heavy molecules in mixtures such as vacuum gas oil (VGO), atmospheric residue, and biomass like vegetable oil. The study explained the preparation of hierarchical zeolite catalysts in detail and described how the catalyst matrix pre-cracks the large molecular weight hydrocarbon feed before it gets into the highly active micropores of the zeolite for selective cracking into the desired products such as gasoline and olefins. It was found that hierarchical catalysts with both microporous and mesoporous structures exhibited a better performance due to the increase in the accessibility and the diffusivity in mesoporous structure and the high catalytic activity in the microporous zeolite.

Previous review studies focused on the catalytic cracking and catalyst deactivation of mainly the conventional FCC processes and catalyst poisons. While some of the reviews have identified the various catalyst poisons and their passivators, to the best of our knowledge, no work in the open literature adequately reviewed the new developments in the use of effective and environmentally friendly passivator. Nickel and vanadium have been identified as the most detrimental poisons of RFCC catalysts as they promote competing reactions such as dehydrogenation and coke formation. Antimony has been used to mitigate the

dehydrogenating effect of nickel. However, it has been found that antimony causes emission issues and is unsafe to handle.

For the first time, this review presents an overview of the current RFCC processes and the various passivators and traps inhibiting the harmful effects of the metals and non-metal contaminants on RFCC catalysts. The advances in the passivation of RFCC catalysts against (i) nickel contaminants: using antimony, rare earth metals, bismuth, and boron as passivators/traps, and those (ii) vanadium contaminants: using tin, rare earth, and basic alkaline compound as passivators/traps were discussed. The associated challenges and future research and development based on (i) design of RFCC catalyst, (ii) replacement of antimony as nickel passivators, (iii) development of RFCC catalyst that incorporates boron as environmentally friendly nickel passivator, and (iv) insight into the passivation of vanadium, were highlighted.

2 RFCC Technologies

Fluid Catalytic Cracking (FCC) is one of the main processes which gives higher operational flexibility due to its ability to process a wide range of feedstocks without sacrificing the feed rate of the unit. FCC is also highly profitable to refiners because the unit alone accounts for around 35 to 50% of the total gasoline pool in the entire refinery, leading to an increased refining margin [19].

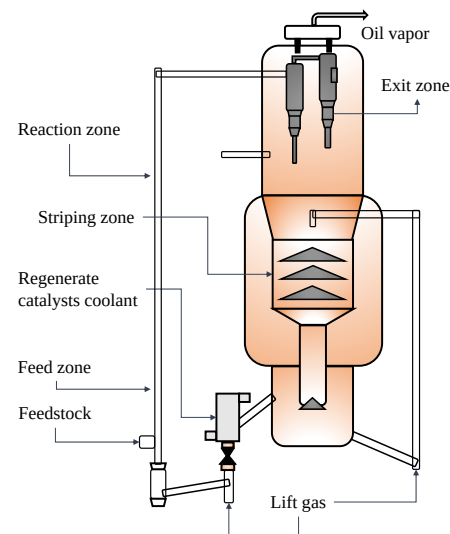
The world's crude supplies have become heavy and dirty due to the decline in the availability of light and sweet crudes. These heavy and dirty crudes contain a high amount of Conradson carbon residue (CCR) and metal poisons such as nickel and vanadium. The high CCR generates a large amount of heat in the regenerator, destroying the cracking catalysts. These current feed characteristics have rendered the conventional FCC unit unfit for use in the modern refinery as it cannot handle the high amount of heat generated due to carbon

151 Table 1: Comparison of Residue Fluid Catalytic Cracking (RFCC) versus Fluid Catalytic Cracking (FCC) [17,20].

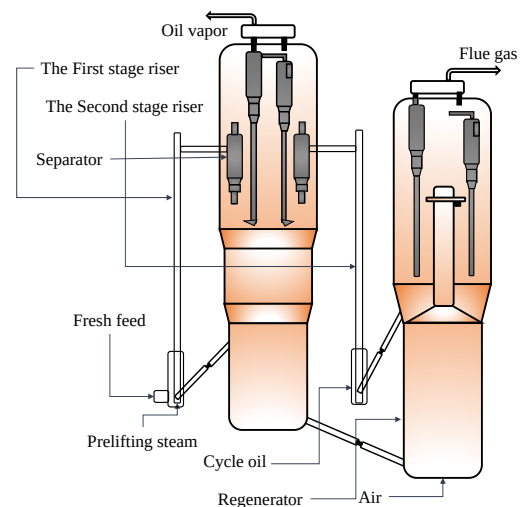
Cracking Unit	Residue Fluid Catalytic Cracking (RFCC)	Fluid Catalytic Cracking (FCC)
Feed Characteristics	- The feedstock is Atmospheric Residue (AR) - Lower API gravity feedstock compared to vacuum gas oil (VGO) (API < 20) - Higher Conradson carbon residue (CCR) (> 4 wt%) - Heavy metal contaminants such as nickel and vanadium are well concentrated in the feedstock	- The feedstock is vacuum gas oil (VGO) and heavy gas oil (HGO) - Higher API gravity and hydrogen content compared to atmospheric residue feedstock (API > 30) - Lower Conradson carbon residue and lower heavy metal contents (< 4 wt%)
Operating Condition	- Contaminant coke is exclusive to RFCC unit operations. - Conradson carbon in the feedstock predominantly winds up as coke. - Heavy metals in the feedstock typically accumulate on the catalyst particles catalysing the formation of coke - A higher temperature in the regenerator - Lower catalyst circulation rate (< 5kg per kg of feedstocks) - Catalyst/oil ratio (5:1)	- Lower coke yield of about 5 wt% on feed compared to residue feedstock with a coke yield higher than 10 wt% on feed. - A lower temperature in the regenerator - Higher catalyst circulation rate (> 5kg per kg of feedstock) - Catalyst/oil ratio (15:1)
Equipment Design	- Two-stage regenerator with catalyst coolers - Combustor-type regenerator for complete combustion, thereby eliminating the cost of installing a CO incinerator	- Single-stage regenerator without catalyst cooler - Bubbling bed regenerator for partial combustion, thereby necessitating the need for a CO incinerator
Catalyst	- The catalyst must have a hierarchical structure to accommodate the large polycyclic aromatic molecules - Catalyst has to be tolerant of the heavy metal contaminants in the feed, notably vanadium and nickel - Catalyst requires a low delta (Δ) coke character to stall the high Δ coke tendencies of the impurities in the feedstock	- Microporous catalyst structure is sufficient to accommodate the molecules of the feedstocks - Catalyst needs a relatively higher delta (Δ) coke to drive the reaction

152 Table 2: Different Catalytic Cracking Licensors and Configurations [21,22]

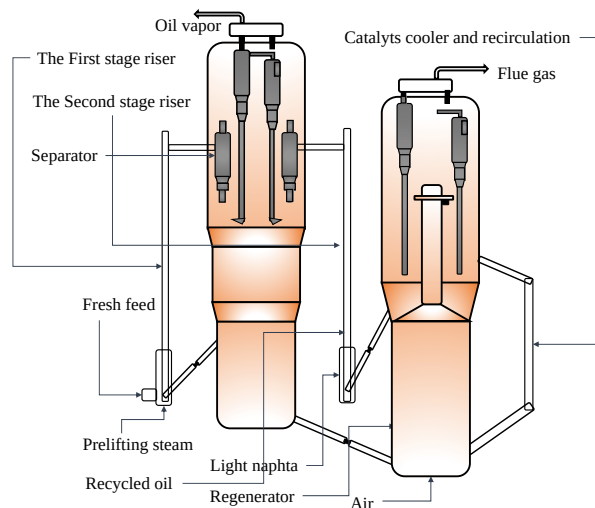
Licensor	Cracking Technology	Configuration	Country
Kellog Brown and Root (KBR)	Orthoflow FCC	Stacked-up	USA
Universal Oil Products (UOP)	FCC and RFCC	Side by side	USA
Axens/Institut Francais du Petrole (IFP)	R2R and RFCC	Side by side	France
Shell Global Solutions International	FCC	Side by side	Netherlands
Foster Wheeler	FCC	Side by side	United Kingdom
Chicago Bridge and Iron Lummus	FCC	Side by side	Netherlands
ExxonMobil Research and Engineering (EMRE)	FCC	Side by side	USA



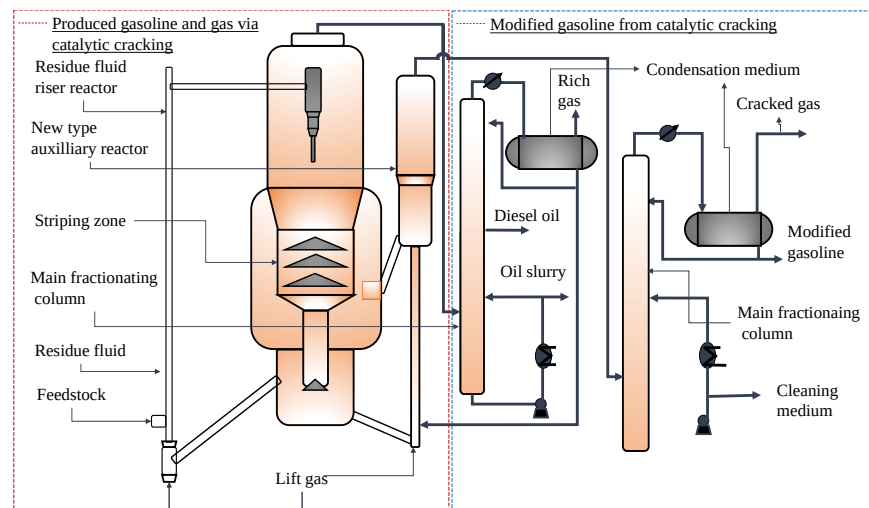
(a)



(b)



(c)



(d)

Fig. 1 Schematic diagrams of residue fluid catalytic cracking (RFCC) technologies: (a) Multi-zone coordinated-controlled, (b) Two-stage riser (conventional), (c) Two-stage riser (for propylene production), (d) Auxiliary riser (Gasoline modification). Technology licensors - China National Petroleum Corporation (CNPC) [23].

residue and the inability of the catalyst to withstand the high level of metal poisons.
Therefore, there is a need for an improved FCC unit to handle the contemporary feedstock

Residue Fluid Catalytic Cracking (RFCC) is a variation of fluid catalytic cracking that has been widely applied in recent years in an attempt to solve the problem of proliferation of heavier and more dirty feedstocks to the conventional FCC unit as a result of the decrease in the world's supply of light and sweet crudes [24]. In this case, the feed stream to the process is the high molecular weight bottom of the atmospheric distillation column, called an atmospheric residue, with high Conradson carbon residue (CCR) and higher contaminant content like metals, nitrogen, and sulfur. Due to the feed stream characteristics, the residue catalytic cracking units require design and optimization changes. The higher levels of CCR in the feed stream led to higher temperatures in the catalyst regeneration step and a lower catalyst circulation rate to keep the reactor at a constant temperature. Reduction in the catalyst/oil ratio, decreasing conversion, and selectivity are some of the adverse effects of high CCR. The RFCC units, however, normally rely on catalyst coolers to avoid these effects [21,24].

Comparison of Residue Fluid Catalytic Cracking (RFCC) versus Fluid Catalytic Cracking (FCC) for feed characteristics, operational condition, equipment design, and catalyst selection [17,20] are presented in Table 1. Due to the high proportion of large molecular weight components, the conventional FCC cannot meet the performance target regarding olefin yield, gasoline production, and lower coke deposition. The higher the amount of high molecular weight compounds, the less the capacity of FCC to meet the performance target. There were several attempts at optimising FCC [25,26]. The FCC unit is the single largest unit in the modern refinery, with several commercial designs with common objectives but different mechanical configurations. The various design configurations of the FCC unit and the major licensors are presented in Table 2.

Improvements have been made to commercial FCC technologies since its inception in 1942, including changes in catalysts, engineering, and facilities. These improvements have increased the reliability of the unit and its capability to process heavier and lower-value feedstocks [27]. The newly improved FCC technologies involve the modification of gasoline, higher liquid yield, higher propylene production, and energy conservation to maximise the output and quality of targeted products and achieve maximum profit efficiency [28]. A brief description and the highlight of the RFCC technologies are presented as follows:

2.1 Multi-zone coordinated-controlled

The multi-zone coordinated-controlled (MZCC) RFCC technology (Fig. 1a) is designed based on a high oil-to-catalyst mixing energy reaction to produce liquid and light petroleum products. The primary benefit of the MZCC RFCC technology is that it increases the yield of light oil and liquids and reduces the formation of coke and dry gas even when processing heavy feedstocks. These benefits are high energy back-mixing for the feeding zone and high effective contact for catalyst and oil. The major disadvantage of this technology is that it leads to an increased level of nickel and iron [23]. These metals deposit on the catalyst and promote unfavourable reactions such as coke formation and dehydrogenation [29]. Some of the technical features of the MZCC RFCC technology [23,28] are:

- High effective contact for oil to the catalyst and high energy back-mixing for the feeding zone
- Rapid separation of oil and catalyst for the exit zone
- Chemical stripping for heavy components for the stripping zone
- Orderly proceeding and plug flow for the reaction zone

2.2 Two-stage riser (conventional)

The conventional Two-stage riser (Fig. 1b) technology comprises two separate riser reactors that can efficiently hinder over-cracking, decrease coke yield and enhance the yield and properties of light oil. The first reactor is for the fresh feedstock and the second reactor is for the entry of cycle oil. The two reactors are in contact with high active regenerant. A two-stage riser removes unwanted product distribution due to an over-long riser and reaction competition in the same reactor between fresh feedstock and cycle oil [30].

The conventional two-stage riser RFCC technology is also designed to produce light petroleum products. Its major advantages come from the short contact time between the catalyst and the hydrocarbon vapour. The brief contact time helps to reduce over-cracking, which reduces dry gas and coke formation. The major shortcoming of this technology is the complexity of the process. Some of the technical features of a Two-stage riser (conventional) [23,30] are:

- Staged reaction: determine reaction condition, and avert competition between cycle oil and fresh feed
- Catalyst relay: improved overall activity due to two-path catalysts circulation, preventing a thermal reaction, and increasing catalytic reaction
- Short reaction duration: lower over-cracking of the target product, decreases coke yield, improves light oil yield, and enhances product distribution
- High catalyst-to-oil ratio: effective increase of the catalyst-to-oil ratio and formation of a new thermal balance due to two path circulation of catalyst

2.3 Two-stage riser (for propylene production)

The two-stage riser for propylene production (Fig. 1c) technology was developed, given the demand for higher propylene production. It is highly selective in propylene and

light oil production. This technology is new and comes with two-stage risers and multiple reaction zones adopted to achieve the following: (i) ideal reaction parameters of integrated feedstock delivery, (ii) high catalyst fluidising density and an appropriate residence time, and (iii) high catalyst-to-oil ratio at low temperature [31].

The Two-stage riser (for propylene production) technology is solely designed to produce propylene. The benefits of the two-stage riser RFCC technology for propylene production are the ideal reaction conditions of integrated feedstock delivery, increased catalyst fluidising density, and a suitable residence time. The major setback of this technology is the increased production of dry gas when processing heavy residue feedstocks. This technology applies to the catalytic cracking of paraffin-based feed to maximise propylene production with the following technical indexes [31] as follow: (i) less than 15 wt% yields of dry gas and coke, (ii) more than 20 wt% of propylene yield and (iii) greater than 93 of gasoline research octane number (RON). Some of the technical features of this technology are:

- Prevention of dry gas formation
- Moderate reaction conditions compared to steam cracking
- Decreased temperature
- Short reaction time
- Increased catalyst-to-oil ratio
- High catalyst fluidising density

2.4 Auxiliary riser (Gasoline modification)

The auxiliary riser RFCC technology (Fig. 1d) is designed for gasoline production and is well known for stable product distribution and reducing olefins in gasoline. The technology has proved to be commercially successful. This technology adds a separate

251 auxiliary reactor to the conventional cracking unit for gasoline modification. Adding the new
252 unit decreases olefins by providing a suitable reaction atmosphere for RFCC, ensuring quality
253 and stable product distribution during the cracking process.

254 The technical features of auxiliary riser technology [23] are:

- 255 • An auxiliary reactor combining a new turbulent and conveying bed on the catalytic
256 cracking unit
- 257 • Auxiliary reactor adopting optimised technological conditions for gasoline
258 modification
- 259 • Flexible regulation for the intensification of reaction conditions leading to a 3 - 6%
260 increased production of propylene

261 However, this technology does not negatively impact the RFCC process when
262 requirements are satisfied. An auxiliary riser for gasoline modification has been applied to
263 five RFCC units in China with favourable outputs [23]. Table 3 depicts the advantages and
264 disadvantages of the various RFCC technologies (multi-zone coordinated-controlled,
265 conventional two-stage riser, two-stage riser for propylene production, and auxiliary riser
266 (gasoline modification). Nonetheless, the successful commercialization of the various RFCC
267 technologies discussed above has shown some major advantages compared to the
268 conventional FCC units. Reduction in gasoline olefins between 20 and 25 v% was attained
269 while maintaining the octane values to satisfy the environmental regulations strictly [23,28].

270 Table 3: The advantages and disadvantages of various RFCC technologies

RFCC Technology	Product	Advantages	Disadvantages	Reference
Multi-zone coordinated-controlled	Liquid and light oil	• It increases the yield of liquid products and light oil • Lowers the yield of dry gas and coke	Increased Nickel and Iron deposits on the catalyst	[30]
Two-stage riser (conventional)	Light oil	• Improves light oil yield and quality, lower over-cracking and decreases coke yield	Complex technology	[23]
Two-stage riser (for propylene production)	Propylene	• Ideal reaction conditions of integrated feedstock delivery, increased catalyst fluidising density, and a suitable residence time	High dry gas production while processing very heavy feedstocks such as residue	[31]
Auxiliary riser	Gasoline	• Stable product distribution and reduction of olefin in gasoline	Expensive and complex technology	[32]

271

Olefin content has become a crucial specification in gasoline because of environmental legislation. Olefin content in gasoline ought to be lower than 30 v%, concurring with the new Chinese national standard, GBI 7930-2009 [33]. The new Chinese standard, however, was achieved with contemporary RFCC technologies. In addition to reducing the olefin level of gasoline, the RFCC technologies have also shown an increase in propylene yield by 3-4% while producing less dry gas and coke. These technologies have helped strengthen the refineries' competitive ability and yielded remarkable economic and social benefits [34].

2.5 Limitations of FCC and the development of RFCC technologies

The fluid catalytic cracking (FCC) process is regarded as the most important refinery process due to its ability to convert the heavy portion of the barrel of crude into more useful petroleum products. In recent times, there has been a quest by refiners to process cheaper crudes. More so, coupled with the decline in the production of light and sweet crudes, the feed to the conventional FCC unit has become heavy and dirty.

Processing heavier feedstocks poses many challenges to the conventional FCC design due to the heavy metal poisons and coke laydown on the catalyst during the cracking reactions. Metal poisons such as nickel, vanadium, iron and sodium deposit on the cracking catalyst and promote a dehydrogenation reaction, thereby shifting the selectivity of the catalyst from the intended products. In addition to fostering undesirable reactions, the metal poisons, especially vanadium, acts to destroy the catalyst structure leading to a loss in activity of the catalyst. A large amount of Conradson carbon residue in the heavy feed leads to a large amount of coke laydown on the catalyst, leading to a large amount of unwanted heat generated during regeneration to restore the catalyst's activity.

These changes cannot be resolved by simple design modifications or changes in the process conditions. Therefore, RFCC technology was developed to address the shortcomings of the conventional FCC process. RFCC operations vary from traditional FCC operations in many significant ways, such as catalyst, operating conditions, feed characteristics, and equipment design, as presented earlier in Table 1. The feed to the RFCC process is the atmospheric residue (AR), the heaviest fraction in a barrel of crude oil and boils at the maximum temperatures. AR contains large quantities of polycyclic aromatics with low API gravity and Conradson carbon residue (CCR) greater than 4wt%. The catalyst for cracking AR feedstock should have features beyond the conventional FCC catalyst, which utilizes vacuum gas oil as feedstock. The catalyst structure needs to accommodate the large polycyclic aromatic molecules in the AR. Such catalyst structure is achieved by incorporating matrix components containing large volumes of pores or creating hierarchical catalyst structures. The catalyst can also be formulated with passivators to take care of the metal poisons present in the heavy residue feedstock. As part of the modification to the equipment design, catalyst coolers are installed in the RFCC process to utilize the large amount of heat generated due to high CCR to produce steam.

3 RFCC Catalyst

A catalyst for cracking residue feedstock must have different features from the catalyst for vacuum gas oil (VGO) feedstock due to the large polycyclic aromatic molecules in residue feedstock. The uniform micropore distribution of zeolites is responsible for their high shape selectivity observed in various reactions. However, this feature limits diffusion when large molecules diffuse in relatively narrow channels. Some active sites are hindered from contacting reactants due to restricted transport processes [35,36]. Consequently, developing new materials with a hierarchical porosity architecture is highly needed. The catalyst for the cracking of residue feedstock has to be designed to tolerate the structure of

the large molecules of residue feedstock because the conventional FCC zeolite catalyst has an average pore diameter of 0.74 nm.

In comparison, the atmospheric residue that serves as feed to the modern RFCC process has molecular constituents with diameters of 1.2 to 15 nm, making it impossible for the large molecules to enter into the zeolite micropores to be cracked directly [37,38]. Designing a catalyst to accommodate the large-diameter residue feedstock is typically accomplished by incorporating a matrix component with a significant pore volume. Several attempts have been made to create RFCC catalysts with large pore volumes by synthesising hierarchical zeolite-Y catalysts using various methods. These methods include the bottom-up and top-down approaches [39–44]. In addition, the catalyst should have a low delta (Δ) coke character to stand the high delta (Δ) coke tendencies of the contaminated residue feedstock as a result of the high amount of Conradson carbon residue (CCR), which is usually greater than 4 wt% in the feed [17]. The low delta coke character of the catalyst is accomplished through specific catalyst design features and targeting an equilibrium catalyst activity more superior than that associated with the VGO catalyst. Finally, the catalyst must tolerate the metals in the feed, particularly vanadium and nickel, as these metals are deposited on the cracking catalyst, destroying the catalyst structure and promoting undesirable reactions. These unwanted reactions include the production of large amounts of H_2 through dehydrogenation and producing large quantities of coke. The large amounts of H_2 are detrimental to the wet gas compressor because it is volume limited. At the same time, a large amount of coke upsets the reactor-regenerator heat balance and affects the main air blower, in addition to temporarily deactivating the catalyst.

Catalysts are continuously being developed for (i) conversion of residue feedstock, (ii) enhanced propylene production, (iii) reduction of olefins in gasoline, and (iv) functional additives. Table 4 depicts some of the zeolite catalysts' physical and chemical indexes to

345 Table 4 Physical and chemical indexes of zeolite-Y catalysts [28].

Physical and Chemical Properties	Reducing olefins in gasoline	Efficient conversion of residue feedstock	Improving propylene production	Functional additives
Al ₂ O ₃ , wt%	≥ 45.0	≥ 48.0	≥ 45.0	≥ 45.0
Na ₂ O, wt%	≤ 0.35	≤ 0.30	≤ 0.30	≤ 0.55
Specific Surface area, m ² /g	≥ 200	≥ 240	≥ 200	≥ 250
Pore volume (droplet method), mL/g	≥ 0.32	≥ 0.35	≥ 0.33	≥ 0.42
Micro-activity (800°C, 4h), kat	≥ 75	≥ 76	≥ 70	-

satisfy the various RFCC processes. However, RFCC catalysts are developed according to the requirements of each refinery to align with operating conditions and the feed characteristics to achieve the desired product with maximum efficiency. For example:

The catalyst for reducing olefins in gasoline: applied to 1.4 million ton/year RFCC unit at Dalian Petrochemical Company in China. The reduction in olefin content of gasoline was recorded to be between 10 - 20% volume ratio, and diesel oil yield increased by over 1.0% mass ratio [28].

The catalyst for residue fluid conversion: applied to a 3.0 million ton/year RFCC unit at Lanzhou Petrochemical Company in China. The residue fluid cracking ability and the overall yield of desired petroleum product both increased by a 0.5% mass ratio [28].

The catalytic cracking catalyst for improved propylene production: applied to 1.4 million ton/year RFCC unit at Lanzhou Petrochemical Company in China. Propylene production increased by 1.0%, while research octane number (RON) increased by more than 1.5 [28].

Functional additives in RFCC: applied to 400,000 ton/year RFCC unit at Lanzhou Petrochemical Company. Propylene yield increased by 1.0%, and the RON of gasoline improved by 0.5 with 5wt% addition of additive [28]

3.1 Catalyst for the Reduction of Olefins in Gasoline

The residue fluid catalytic cracking (RFCC) gasoline usually contains 20 – 40 wt% of olefins [45–47]. The presence of olefins in RFCC gasoline is advantageous because it helps to increase the gasoline octane number. However, the high content of olefinic compounds affects the engine exhaust emissions and the stability of the fuel under storage conditions. Olefins react with atmospheric oxygen leading to oxidation and polymerisation products,

forming gum in the gasoline tank [48,49]. According to European Standard EN228, the olefin content in gasoline is currently limited to 18 vol% [50].

Several attempts have been made to reduce olefins by hydrogenation [45,51–53] to improve the stability of RFCC gasoline. However, the process leads to an undesirable decrease in the gasoline octane, resulting from converting high-octane olefins into low-octane paraffin [51,52,54,55]. Recently, Dubkov et al. [46] used nitrous oxide (N_2O), a greenhouse gas, to oxidise light-catalytic cracked gasoline. The treatment of the gasoline with N_2O was performed without catalyst in the liquid phase at 200 – 250°C and elevated pressure of 50-70 atm. The findings suggest that N_2O selectively oxidises olefinic compounds in the gasoline to form carbonyl products and molecular nitrogen without affecting the gasoline components, making N_2O a good candidate for olefin reduction in gasoline.

In 1996, the Kashima refinery in Japan revamped its FCC unit by modifying it to a new UOP RFCC. The modifications include (a) a catalyst cooler, (b) an expanded air blower and gas compressor, (c) improved atomisation feed nozzles, and (d) a design with a short contact time (SCT) riser with a cyclone riser directly connected to riser termination. The revamp allowed the FCC unit to increase throughput from 20,000 to 28,500 bpsd with a higher intake of residue feedstocks. Modified unit and adjusted feed rate and composition produced gasoline with high olefin content. A switch to a new catalyst by Akzo Nobel was designed to reduce olefin content while maintaining the RON of the gasoline. The new catalyst design for controlling gasoline olefins reduced olefins concentration in gasoline by more than 5 v% without a significant RON decrease [56].

3.2 Residue fluid conversion catalyst

Due to the increasing scarcity of crude oil resources, the feed to the residue fluid catalytic cracking unit has become heavier and laden with heavy metals, for example,

vanadium and nickel, which are harmful to the life of the catalyst. For this reason, superior technology has been developed to improve heavy metal pollution and render the catalyst matrix less active to the effects of nickel and/or vanadium [30,37,57].

Zeolite-Y, with a uniform microporous structure and an inlet diameter of 0.74 nm, is the main active component of the catalyst used in the conventional FCC process [38,58,59]. However, the large molecular diameter (1.2 to 15 nm) of atmospheric residue, which serves as feed to the modern RFCC process [37], makes it nearly impossible for the large molecules in heavy oil to access the micropores of the zeolite to be cracked directly [38,60,61]. Therefore, it is desired that RFCC catalyst for processing heavy residue feedstocks must have improved diffusion abilities, enhanced acid site accessibilities and tolerance to metal poisons [62–67]. Much research has been undertaken to introduce meso- and macroporosities into zeolite crystals through developing hierarchical zeolite materials and synthesising metal-resistant catalysts [68,69].

Recently, attention has been paid to core-shell structured catalysts because they can provide concrete reaction pathways and improve the synergy of many active sites on the formulated catalyst [70,71]. For this reason, many core-shell structured composites that employ zeolite crystal as core and mesoporous matrix as shell are developed to process residue feedstocks more efficiently [62,63,65–67]. This core-shell composite may improve mass transfer limitations, but still, there are challenges concerning metals poisoning.

3.3 Catalytic cracking catalyst for improving propylene production

Propylene is an important petrochemical feedstock for producing products such as acrylonitrile, butyl alcohol, cumene, phenol, polypropylene, and propylene oxide [72]. Propylene production is broadly categorised into steam and catalytic cracking paths [73]. The steam cracking path is the highest energy-consuming process in the petrochemical industry.

The type of feedstock and operating conditions considerably impact propylene production in the steam cracking process. The reaction temperature can reach 800°C for steam cracking by releasing large amounts of CO₂. The catalytic cracking path operates at a lower temperature of around 500°C and has higher propylene production [74].

RFCC technology for higher propylene production has drawn much attention because of its low-cost and abundant feedstock compared with conventional steam cracking technology [75]. A proper catalyst design and preparation are required for optimal propylene production. Higher propylene production can be achieved by adjusting the acid quantity, strength and type of RFCC catalysts [73,76].

To improve propylene production, Liu et al. [77] investigated an RFCC catalyst for maximising propylene production by mixing the modified NaY/kaolin composite microspheres and additive microspheres containing ZSM-5. The characterisation results showed that the catalyst contained higher meso- and macro-pores, had higher acid sites than the reference catalyst, and increased propylene production by approximately 1%. In a similar attempt, Gao et al. [78] investigated phosphorus-modified ZSM-5 prepared via hydrothermal dispersion in the cracking of light diesel for maximising propylene production. The result showed that more pronounced cracking activity was obtained on the catalyst containing phosphorus, and propylene production increased by 3% compared to the reference ZSM-5 catalyst. The increase in propylene production can be attributed to the higher amount of phosphorus fixed on the zeolite prepared by hydrothermal dispersion and the higher residual ratio of crystallinity.

Numerous studies have also investigated the suitability of hierarchical ZSM-5 for improving propylene production [79–82]. Hierarchical ZSM-5 has shown excellent shape-selective catalytic activity owing to its porous structure. It also offers good hydrothermal stability and strong resistance to coking. The stability of the catalyst is attributed to the

aluminium content and the shape selectivity given by the microporous network of the zeolite cavities. In contrast, the coke tolerance is attributed to the hierarchical nature of the catalyst, which makes it difficult for the catalyst pores to be blocked [79–82].

Bulatov and Jirnov [83] investigated the conversion of heavy feed at various concentrations of ZSM-5 additive between 0 and 40 wt % at a catalyst-to-oil ratio of 28, a riser outlet pressure, temperature and contact time of 0.0739 Mpa, 566°C, and 1.5 seconds respectively. An increase in the amount of ZSM-5 to significantly high concentrations, up to 40 wt%, was found to have little effect on propylene production. The propylene yield increased by around 10% of ZSM-5 crystal concentration in the catalyst inventory. It was concluded that the decrease in the effectiveness of ZSM-5 at higher concentrations transpired mainly due to the exhaustion of the gasoline olefin precursor.

Konno et al. [84] studied the effect of zeolite crystal size on ZSM-5 (Si/Al = 150) catalytic stability in n-hexane under atmospheric pressure at temperatures ranging between 550°C and 650°C. The findings showed that the Si/Al ratio of the zeolite affected the product selectivity and production of propylene. In the study, ZSM-5 with different Si/Al ratios of 50, 150 and 300 were prepared via hydrothermal synthesis in a water/surfactant/organic solvent. It was found that ZSM-5 with Si/Al of 150 exhibited the highest propylene production of 50.4% at 650°C because excessive reactions such as the consumption of light olefins and BTX formation were suppressed by the low acid density (high Si/Al ratio). It was concluded that applying ZSM-5 zeolite (Si/Al = 150) to the cracking reaction at a high reaction temperature of 650°C improved propylene yield.

3.4 Functional additives for catalytic cracking

Functional additives are catalyst materials added separately to the base RFCC catalyst, blended with the catalyst during formulation or even introduced into the feedstock.

The main benefits of these additives are to increase gasoline octane number, increase propylene production, minimise metals contamination of the catalyst and reduce the number of pollutants emitted from the RFCC regenerator [85–89].

The combustion of coke in the RFCC regenerator produces several atmospheric pollutants that need to be controlled. These pollutants include carbon monoxide (CO), sulfur oxides (SO_x), nitrogen oxides (NO_x), and particulate matter (PM). However, additives are customised with particular properties based on requirements without altering product distribution.

A practical approach toward NO_x emission mitigation from fluid catalytic cracking units was investigated by reviewing options for cost-efficient and pragmatic emissions mitigation using optimal amounts of precious metals while evaluating the potential benefits of typical promoter dopant packages [90]. The findings suggest that at the present catalyst development level, the refinery is no longer forced to make a promoter selection based on preconceived notions regarding precious metal activity but can rather make decisions based on the best total cost financial impact to the operation without measurable loss of the CO/NO_x emission selectivity.

Mo et al. [91] studied SO_x reduction additives for partial burn FCC by employing a commercially available SO_x reduction additive and an additive developed by JM Savannah. The authors found that among other SO_x reduction technologies, the use of additives in FCC units provides a simple and cost-effective method and has been found essential by most refiners. It was also pointed out that the reduction of SO_x by additives was achieved via SO_x capture in the regenerator and the release of H₂S after being transferred to the riser. Chester et al. [92] carried out a study on CO combustion promoter additives in FCC regenerator unit by employing CO promoter containing various noble metals (platinum (Pt), Palladium (Pd), Iridium (Ir), Rhodium (Rh), Os (Os) and Rubidium (Ru)). The findings

suggest that when 10 ppm of Pt, Pd and Rh were used, oxidation activity (CO_2/CO) were 49, 36 and 26, respectively and when 5 ppm of Ir, Ru and Os were used, CO_2/CO were 43, 8.6 and 2.2 respectively. It was concluded that CO combustion promoters containing Pt are the best at oxidising CO to CO_2 , meaning that CO promoters containing Pt should be employed when controlling the burning problem in the FCC regenerator after considering the cost.

3.5 Reaction pathways of the RFCC process

The pathways of the reduction of olefins in gasoline are via (i) aromatisation of olefins in the absence of hydrogen and (ii) hydroisomerisation and aromatisation of olefins in an excess hydrogen atmosphere (Fig. 2). Fig. 2 (a) shows the aromatisation of olefins in the absence of hydrogen. Here, the initial intermediates are dienes, and the subsequent intermediates are cyclo-olefins. Based on the viewpoint of the hydrocarbon, the dienes conversion into cyclo-olefins is more likely than the direct conversion of n-olefins into cyclo-olefins. The phenomenon, however, suggests that the formation of cyclo-olefins from dienes is too fast to be tracked. Fig. 2 (b) depicts the hydroisomerisation and aromatisation of olefins in an excess hydrogen atmosphere as convened in FCC gasoline hydro-upgrading.

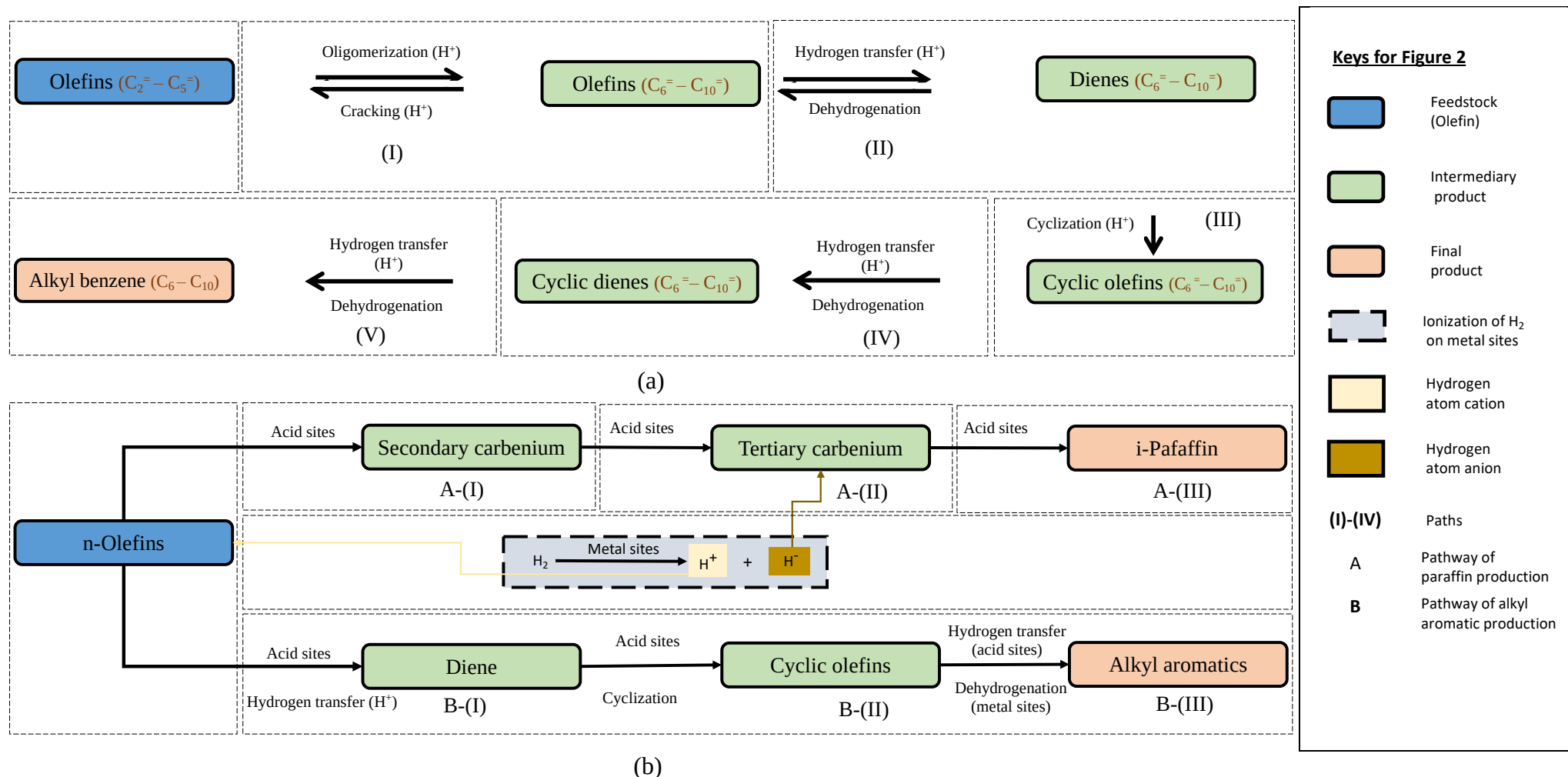


Fig. 2 Reaction pathways for the RFCC processes for Reduction of Olefins in Gasoline (a) aromatization of olefins over HZSM-5 and Ga/HZSM-5 catalysts in the absence of hydrogen and (b) hydroisomerization and aromatization of olefins over Ni-Mo/modified HZSM-5 + Al_2O_3 catalyst in the presence of hydrogen. Reproduced with permission from [93] Copyright {2009} American Chemical Society.

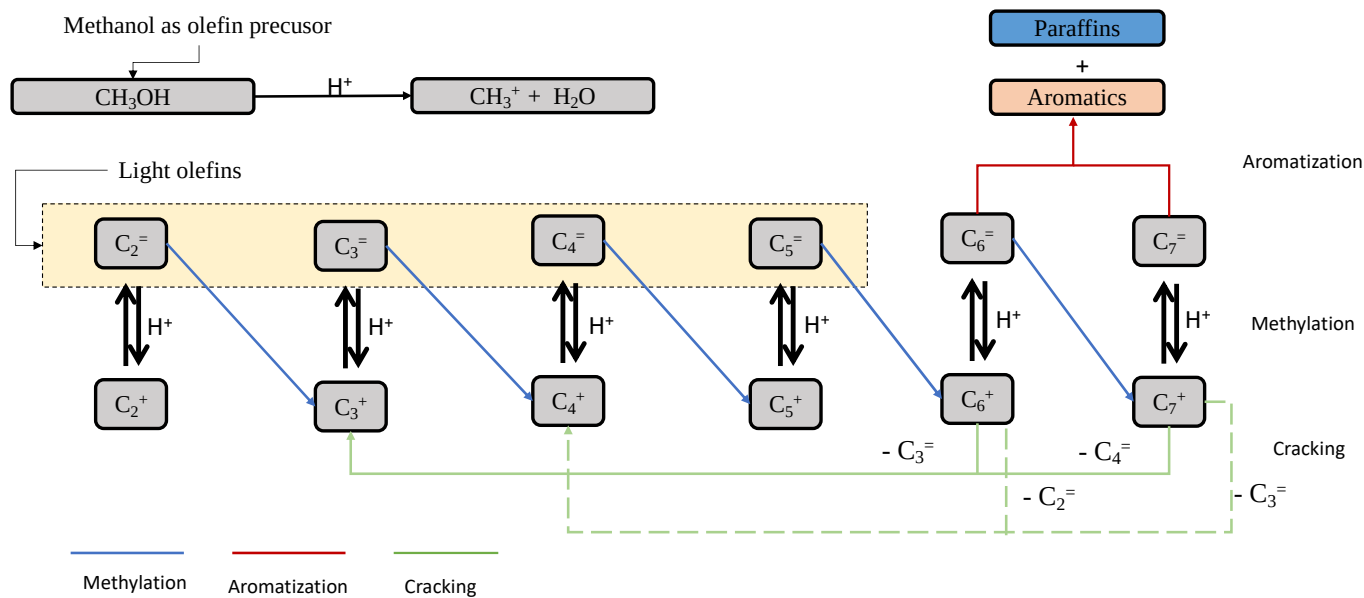
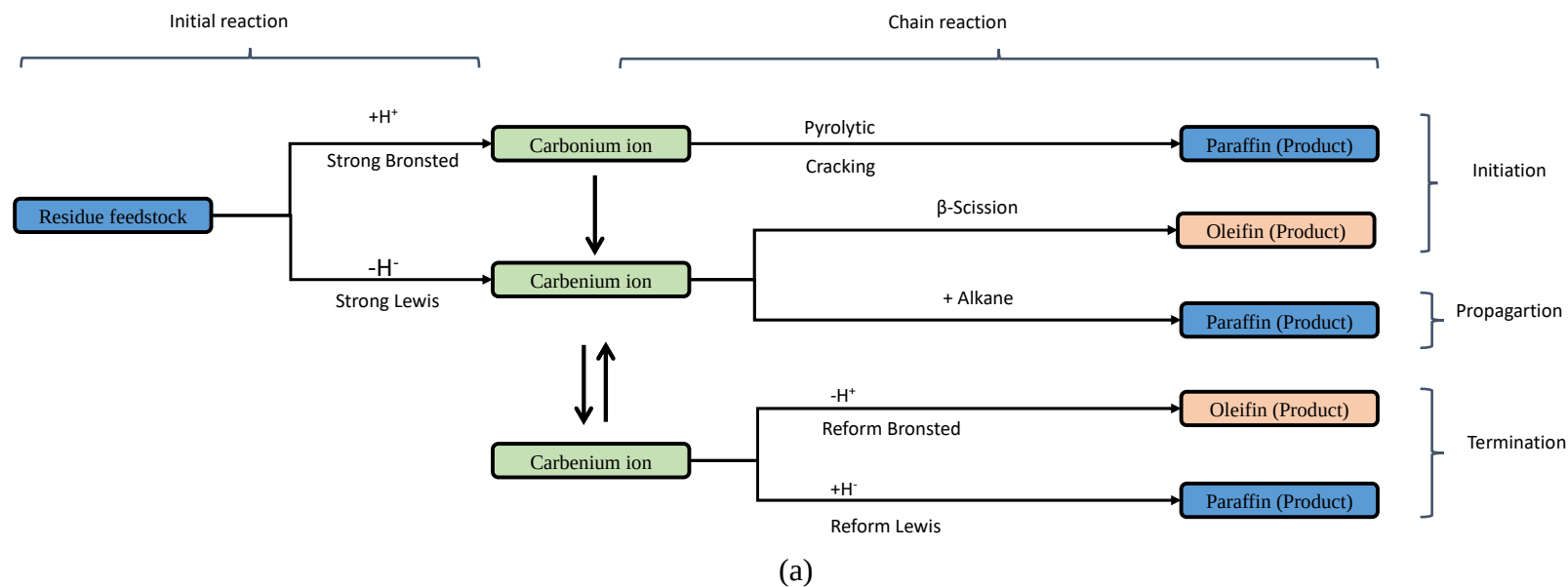


Fig. 3 Reaction pathways for the RFCC processes for (a) residue conversion catalysts. Reproduced from [94] Springer nature {2013}, (b) improving polypropylene production. Reproduced with permission from [95] copyright {1980} Elsevier.

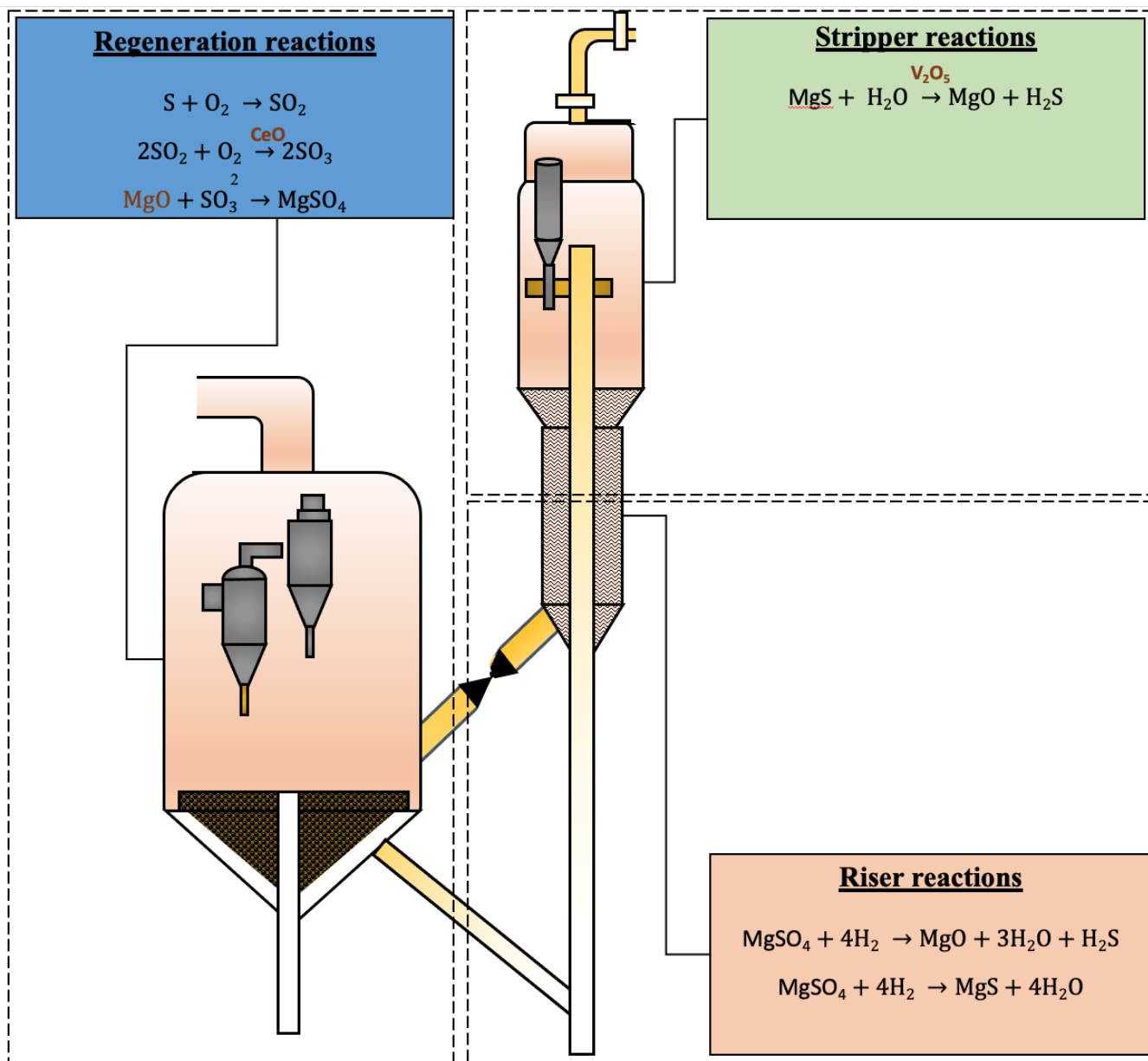


Fig. 4 Reaction pathways using functional additives in RFCC. Reproduced with permission from [96] copyright {2017} Elsevier.

First is the absorption of olefin molecules on the acid sites of the catalyst (bifunctional). A portion of the olefin molecules is transformed into carbenium ions via a combination with H^+ ions spilt over from the metal sites. However, the residual olefins are converted into dienes via a hydrogen transfer on the acid sites. Afterwards, the carbenium ions formed are converted to i-paraffins by isomerisation, rearrangement, and hydrogen spill-over. Simultaneously, the dienes are transformed into aromatics and cyclo-olefins by cyclisation, hydrogen transfer, and hydrogen transfer/dehydrogenation [97].

Catalytic cracking is catalysed by Bronsted and Lewis acid sites of zeolites and involves breaking up of C-C bond forming of carbonium and carbenium ions. Fig. 3 (a) illustrates the reaction pathway of cracking residue feedstock via carbonium and carbenium ion intermediates. It includes an initiation on any acid Bronsted site by protonation or the Lewis site hydrogen abstraction. Subsequently, propagation follows with hydride transfer and cracking via β -scission. Termination reactions are followed by the reaction of a carbenium ion as follows:

- (i) a Bronsted base forms a paraffin.
- (ii) a Lewis base or Bronsted acid site creates paraffin and reforms a Lewis acid.
- (iii) a hydride ion derived from coke forms a paraffin.

Likewise, initiation and cracking can arise via paraffin's reaction and a Bronsted acid site creating a carbonium ion, which can subsequently react via β -scission to a carbenium ion and smaller paraffin or undertake H_2 elimination to a large carbenium ion. Fig. 3 (b) illustrates the main reaction pathway forming light ($C_2 - C_4$) olefins from methanol conversion at steady-state operation over an H-ZSM-5 catalyst. After the initial olefins are formed during the induction period, the olefins are consecutively methylated to form higher olefin homologues, which subsequently crack to form lighter olefins such as ethylene and

propylene. Hydrogen transfer and cyclisation reactions lead to the formation of paraffins and aromatics as end products [98,99].

The reaction pathway of the RFCC process using functional additives (Fig. 4) shows the mechanism of the SO_x removal additive [96]. Typically, a SO_x additive comprises multiple active constituents, including vanadium oxide, cerium oxide, and magnesium oxide. The relative quantities of the main components are optimised for the best performance. Sulfur is oxidised to form sulfur dioxide through coke combustion in the regenerator. In the first step, sulfur dioxide is oxidised to sulfur trioxide, which reacts with the magnesium oxide (in the additive), forming magnesium sulfate. In the second step, the additive gets to the riser, and the magnesium sulfate is reduced by reducing gases such as hydrogen, methane and other volatile organic compounds to regenerate the magnesium oxide and produce magnesium sulfide. Finally, in the stripper, the magnesium sulfide undergoes hydrolysis in the presence of vanadium pentoxide and steam to form magnesium oxide. An efficient SO_x additive should (i) oxidise sulfur dioxide to sulfur trioxide, (ii) form a stable magnesium oxide, and (iii) effectively liberate sulfate species to regenerate the activity of the additive. Spinel, a mineral composed of magnesium aluminium oxide (MgAl₂O₄) and hydrotalcite, a layered double hydroxide, is the commonly used commercial SO_x additive [96].

4 RFCC Catalyst Deactivation

Refineries frequently find an opportunity to buy low-value or lower-cost crude feedstocks to reduce operating costs. Although these opportunity feeds can help improve profit margins, they usually have serious consequences. These consequences include the destruction of catalysts and the production of a large amount of hydrogen and coke at the expense of gasoline due to unspecified concentrations of impurities in the feed [14,100–102]. Therefore, it is vital to be aware of such impurities to avoid surprises, such as catalyst

deactivation and high coke deposition, which can cause indiscriminate product distribution and shorten the cycle length of cracking catalysts.

Modern RFCC cracking catalyst deactivates rapidly at temperatures above 730-745°C due to hydrothermal deactivation caused by high temperatures and a steam environment. The high temperature is responsible for the catalyst's sintering (melts and collapses) [103].

Cracking catalyst also deactivates by the sodium and vanadium that accumulates on the catalyst. Sodium neutralises some of the acid sites within the zeolite structure, rendering those acid sites permanently inactive. Sodium poisoning is usually caused by sodium chloride that enters the crude unit during a desalter upset. However, it is well controlled in most refineries using double desalters in the crude unit. The fresh catalyst typically has 2000 to 3000 ppm sodium left from manufacturing. Additive sodium beyond 5000 ppm on an equilibrium catalyst can cause a rapid decline in catalyst activity [104].

Vanadium, when deposited on the catalyst particle surface, oxidises in the regenerator to V_2O_5 , which in the presence of steam becomes mobile vanadic acid (H_3VO_4) that migrates throughout the catalyst particle and then destroys the zeolite silica-alumina framework [105,106]. Vanadium is organically bound within organometallic molecules and tends to concentrate in the heaviest fraction of the feedstock. Modern vanadium trapping technology allows concentrations of up to 7000 ppm vanadium on equilibrium catalyst to be tolerated while maintaining suitable catalyst activity [107–109].

Nickel is also organically bound within organometallic molecules and tends to concentrate in the heaviest fraction of the feedstock. Nickel is a dehydrogenation agent and catalyses the formation of H_2 and coke. However, nickel deposits on the surface of the catalyst particle and, once deposited, tend to be stationary. The signature symptom of nickel poisoning is high H_2 formation, which results in high dry gas production. The formation of a high amount of hydrogen can be countered by modern nickel trapping technology, lift gas

586 Table 5: Effects of Non-Metal and Metal Contaminants on RFCC Catalyst.

Types	Contaminants	Effects on RFCC Catalyst	Causes	References
Non-metals	Nitrogen	• Blocks catalyst pores • Increase coke formation • Poisons acid sites • Reduces activity	Adsorption of basic nitrogen compounds (pyridine, quinolones, acridine and phenanthridine) on the active sites of the catalyst	[100,110–113]
	Sulfur	• Decreases activity • Increase coke formation • Destroys acid sites	Formation of thiophenes that neutralises the Bronsted acid sites	[114–116]
Metals	Iron	• Blocks catalyst pores • Obstructs reactants' access to active sites	Improper desalting of feedstock to RFCC unit, the resulting product of corrosion from storage equipment	[117,118]
	Sodium	• Neutralisation of acid sites • Destruction of the zeolite framework	Lowering the melting point of the catalyst, thereby making it susceptible to sintering at high temperature	[104–106]
	Nickel	• Nickel does not affect the catalyst structure but promotes undesirable reactions and unwanted products	Covering the catalyst surface with nickel makes the surface thicker and forms crystalline sulfides that migrate into the porous structure	[102,119]
	Vanadium	• Surface area loss, • Dealumination • Unit cell size shrinkage • Acid sites destruction	Hydrolysis of silica-alumina framework by volatile and mobile vanadic acid	[15,101,120–122]

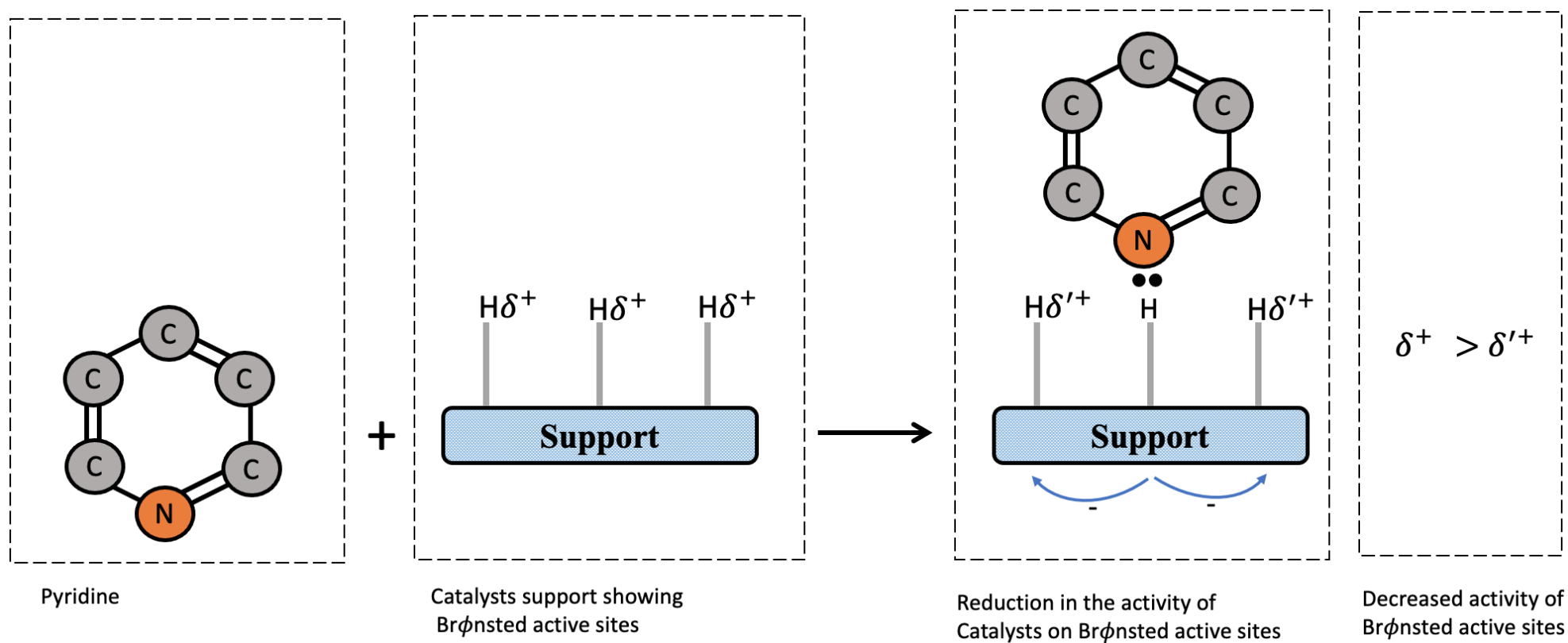
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and antimony to passivate the dehydrogenation impact of the nickel. New nickel will deposit on top of the old nickel, rendering the old nickel inactive. H₂ production tends to rise rapidly with the initial nickel deposits but then tapers off with further increases in nickel. Incremental nickel levels above 8000 ppm produce little incremental H₂ [17]. Table 5 summarises the effects and causes of the various non-metals and metals contaminants on RFCC catalysts.

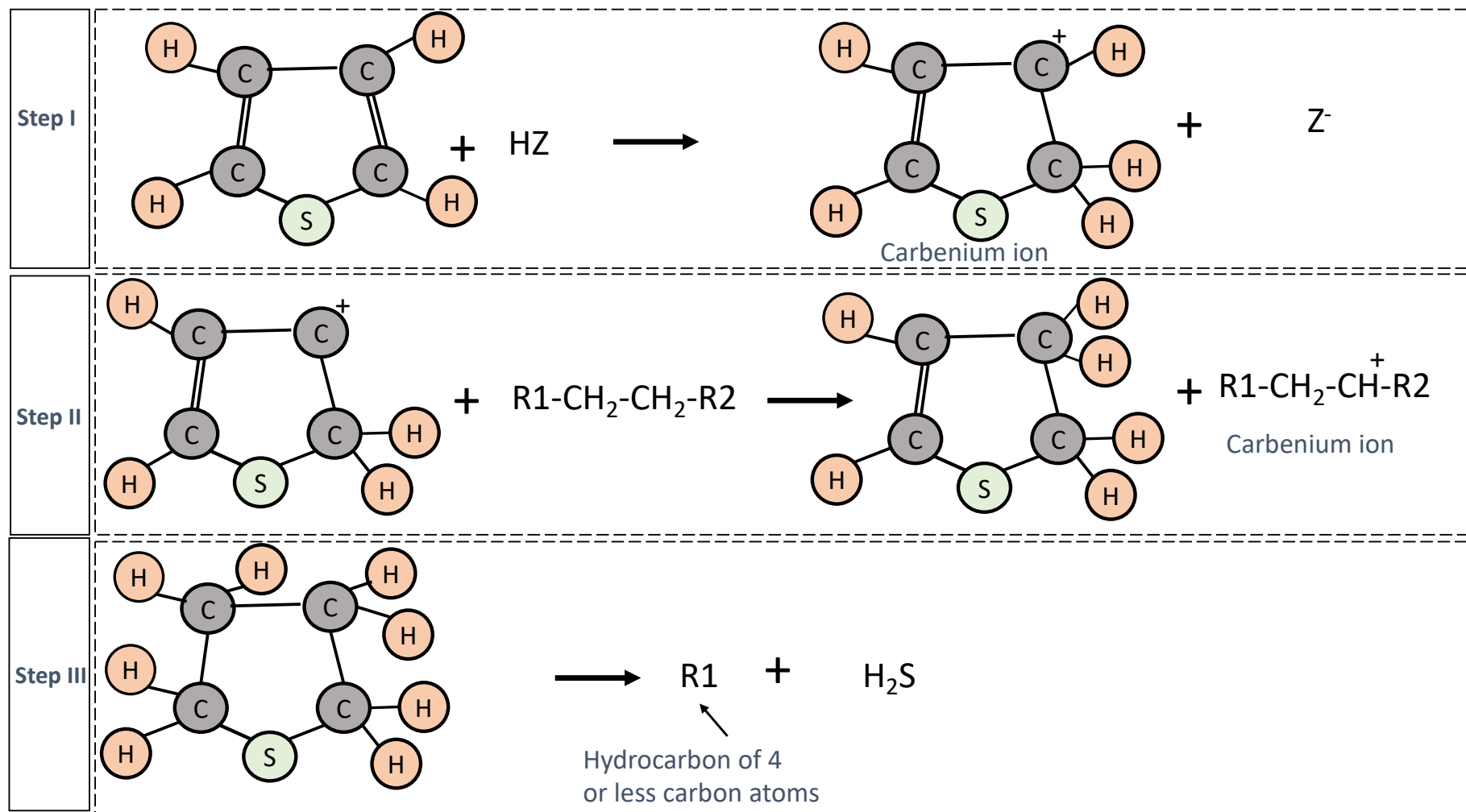
4.1 Nitrogen poisoning

Over the years, the poisoning effect of basic nitrogen has been extensively published [100,111,112,123–125]. Recently, there has been a shift in crude oil supply towards heavier crudes. Heavy crude contains a high concentration of nitrogen (8000 ppm and above) compared to lighter crude (0 – 5000 ppm). This shift necessitates the need to devise a means of dealing with this sharp increase in nitrogen content in the processing of the heavy crude feedstock. High-nitrogen crudes have nitrogen content greater than 0.25 wt%, and nitrogen-poor crudes with nitrogen content lower than 0.25 wt% [126]. In the lighter fractions of crude oil, nitrogen is mainly in the form of basic compounds (pyridines, quinolines and benzoquinolines). On the contrary, the nitrogen sources in heavy crude oil are in the form of non-basic compounds such as indoles, pyrroles and carbazoles [110].

Two mechanisms have been used to explain the retarding effects of basic nitrogen compounds [111,127]. First, the basic nitrogen compounds reversibly absorb on the Brønsted or Lewis acid centres and then poison the acid centres hindering the hydrocarbon cracking ability of the FCC catalyst. Second, because of their big size and aromatic nature, the basic nitrogen compounds may be the precursors of coke. Li et al. [113] found that the retarding effects of non-basic compounds and condensed aromatics in coker gas oil catalytic cracking are less serious than those of basic nitrogen compounds. Fig. 5 depicts the proposed mechanism for the inductive partial poisoning effect of nitrogen on neighbouring acid sites



612 Fig. 5 Mechanisms of the retarding effects of basic nitrogen compounds on the Brønsted acid centres. Reproduced with permission from [12]
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614 Fig. 6 Mechanism of thiophene hydrogenation and H₂S production. Reproduced with permission from [93,128,129] Copyright {2006}. Elsevier

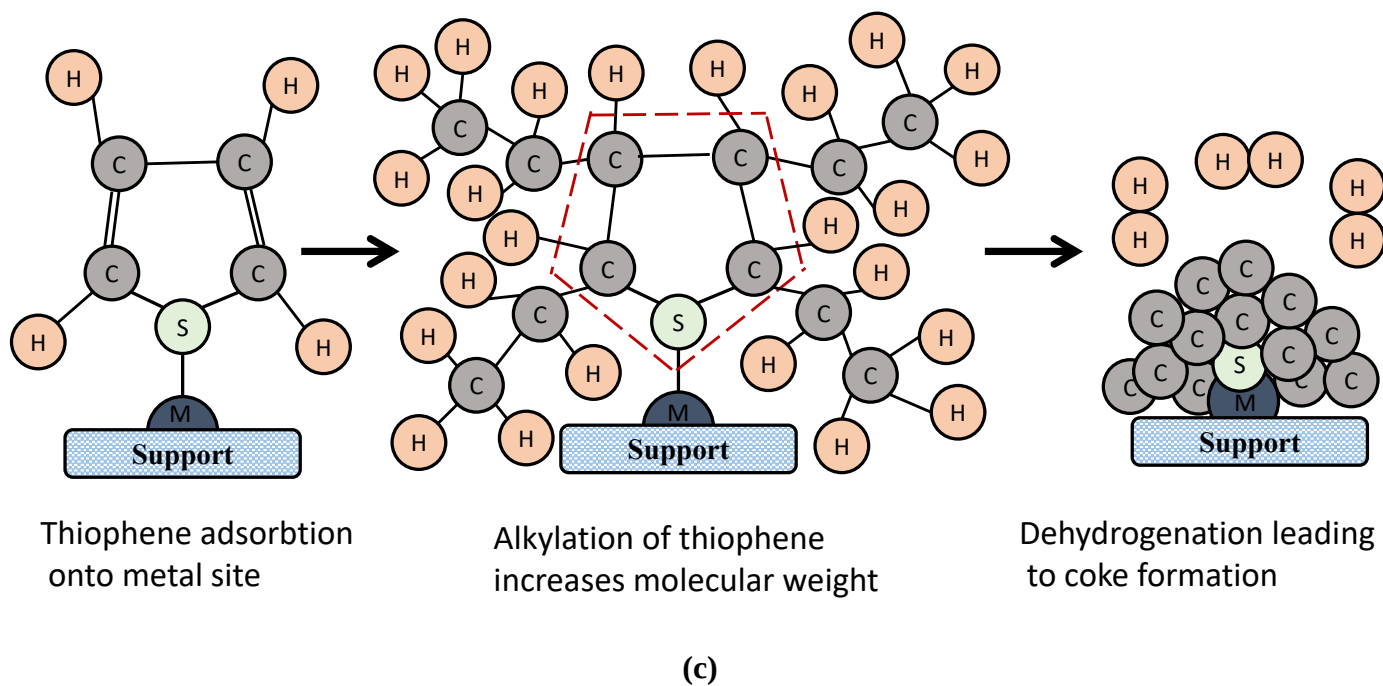
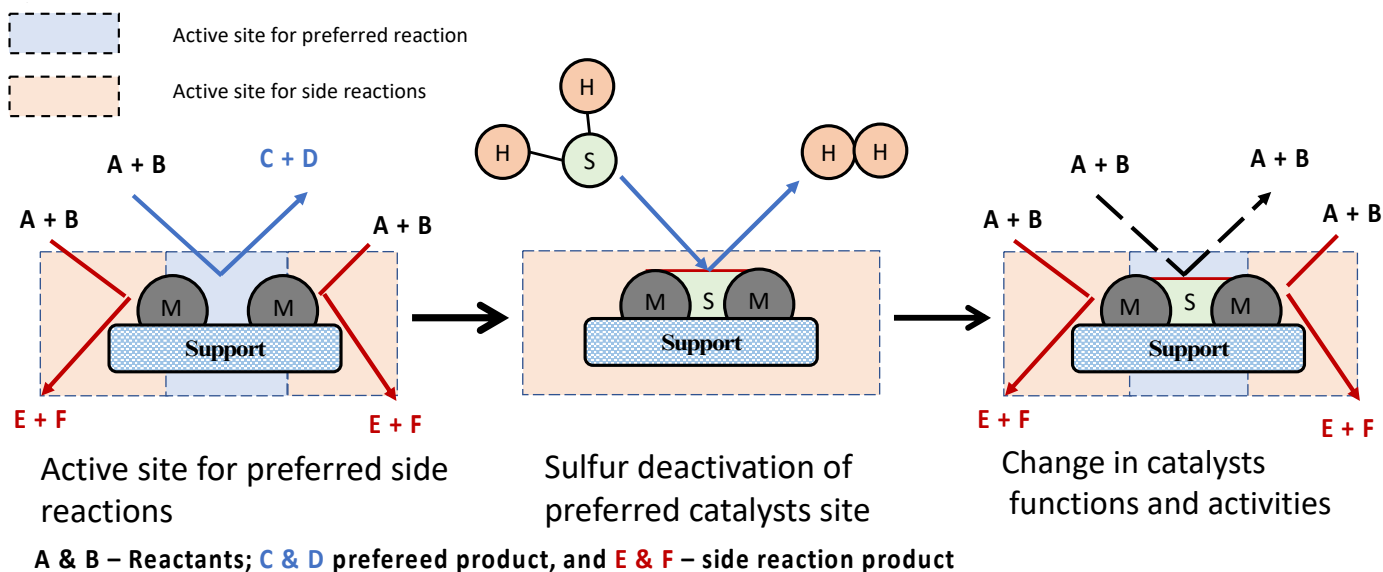
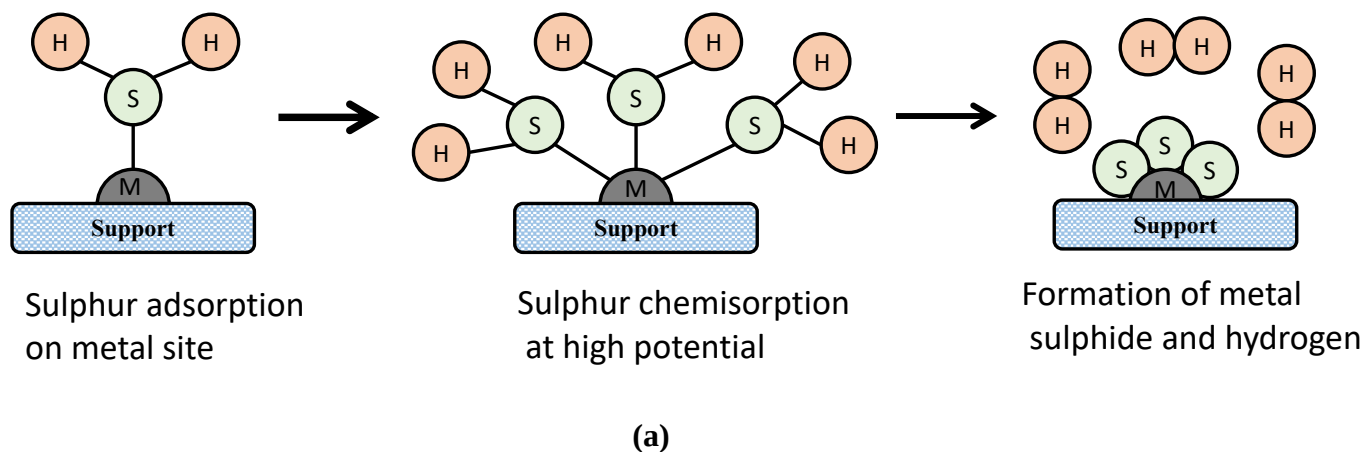


Fig. 7 The three major sulphur deactivation mechanisms of catalysts. (a) Sulfidation, (b) Alteration, and (c) Coke formation. Reproduced with permission from [93,129] Copyright {2019} Elsevier.

by strong basic nitrogen compounds such as 2,6-dimethylpyridine relative to quinoline and pyridine. 2,6-Dimethylpyridine seemed to induce stronger deactivation than quinoline and 3-methylpyridine, which are less basic. However, due to steric constraints, 2,6-dimethylpyridine is only adsorbed on the reactive Bronsted acid sites, which could also explain the higher poisoning strength [12].

4.2 Sulfur poisoning

Sulfur is regarded as the amplest heteroatom in crude oil. In most cases, the concentration of sulfur in crude increases with a decrease in API gravity, which means that the heavier the crude, the greater the concentration of sulfur [115,116,130]. RFCC catalyst sulfur comes mainly from the cracking of long alkyl chain thiophenes. The mechanisms of thiophene hydrogenation and H₂S production are presented in Fig. 6.

Alkylated thiophenes, benzothiophenes and dibenzothiophenes are the major sulfur compounds in typical RFCC feeds [12,45]. The deactivation mechanism of RFCC catalyst by sulfur molecules occurs through interaction with acid sites and condensation reactions of their unsaturated rings leading to coke formation [12,131]. The coke formed can poison the acid sites or block the pore structure.

Sulfur is present in hydrocarbon feedstock in the form of hydrogen sulfide, sulfur dioxide, thiophene, and its derivatives. These sulfur compounds contribute significantly to catalyst poisoning. However, it is thought that the poisoning mechanisms of sulfur compounds are different and lead to a reduction in catalyst performance through different deactivation pathways. Fig. 7 illustrates the three major sulfur deactivation mechanisms commonly encountered in hydrocarbon processing. The mechanisms include the sulfidation, alteration and coke formation mechanisms[129].

The sulfidation step (Fig. 7a) involves the direct interactions between sulfur and catalyst active sites. These include the adsorption of sulfur compounds and the formation of sulfides. Catalysts exposed to low H₂S concentrations are typically deactivated via the adsorption of sulfur onto the catalyst surfaces, thereby blocking off access of the reactants to the active metal sites. The adsorption is usually reversible, and catalytic activity can be regenerated by raising the temperature or lowering the feed sulfur content to desorb the sulfur from the poisoned surfaces. At higher H₂S concentrations, sulfur's chemical potential increases, favouring the chemisorption of sulfur into the catalyst material, leading to subsequent sulfidation reactions to give metal sulfides. The sulfidation reactions of transition metal active sites are generally irreversible and can result in permanent loss in activity of the catalyst, as demonstrated in Fig. 7b.

In the alteration step (Fig. 7b), sulfur compounds interact with the active sites of the catalyst and deactivate the preferred catalyst sites, thereby leading to a change in catalyst functions and activities. This poisoning can alter the functionality of catalysts by affecting the affinity, selectivity and reaction pathways of adsorbed reactants on the surface of the metal sites. The coke formation step (Fig. 7c) involves the role of sulfur compounds in the growth and deposition of carbonaceous coke on the catalyst, thereby reducing the activity of the catalyst. Various studies have suggested that coke formation is accelerated for sulphur-poisoned catalysts due to preferential interactions of alkyl radicals with the sulphur atoms to form R-S species. The R-S species then rapidly dehydrogenate to give indirectly-bonded coke deposits on the surface of the catalysts' active sites, which are very stable [132,133].

4.3 Iron poisoning

Iron deposits on the RFCC catalyst as organic iron from the feed or inorganic iron from equipment corrosion. When deposited on the RFCC catalyst, this Fe promotes

undesirable dehydrogenation reaction and catalyst circulation problems due to the formation of nodules on the catalyst surface. Rainer et al. [134] suggested that when the RFCC catalyst was contaminated with Fe, analysis using Atomic Force Microscopy (AFM) indicated a bumpy and nodulated appearance on the surface of the RFCC catalyst, leading to catalyst circulation problems. Similarly, Psarras et al. [135] noted that as the level of iron increases on the surface of the RFCC catalyst, the sintering phenomenon results in the imposition of diffusional barriers to the migration of heavy molecules of a hydrocarbon feedstock to the active site of the catalyst and this led to poor performance of the RFCC unit. The relationship between RFCC unit performance and iron contamination is well-established in the literature [136,137]. For example, it has been well-established that iron deposits on FCC catalysts form a dense layer on the catalysts' surface, and the dense layer stops reactants from diffusing to the inner structures of the catalyst. This causes adverse effects on the catalyst's ability to convert heavy oil and could greatly reduce gasoline yield while increasing yields of dry gas and coke. The reaction performance of the catalyst would be severely affected when the iron content on the equilibrium catalyst exceeds 8000 ppm [29,138].

4.4 Sodium poisoning

Sodium is an important heteroatom that is part of the hydrocarbon structure. It may also be from salty water. It is a deleterious cause of poisoning to watch out for in the RFCC feed. Sodium comes from various sources, including naturally contaminated oil feedstocks, a remnant from oil extraction due to a faulty desalter, or a residual material from catalyst preparation. Sodium is well known to cause the destruction of zeolite framework, participate in the ionic exchange and reduce the amount of Bronsted acid sites [104,105,139]. Oliveira et al. [140] and Xu et al. [141] reported that Na and V could interact synergistically with steam

to yield NaOH. The NaOH formed migrates into the zeolite structure and readily dissolves crystalline components, negatively affecting the catalyst's activity and selectivity.

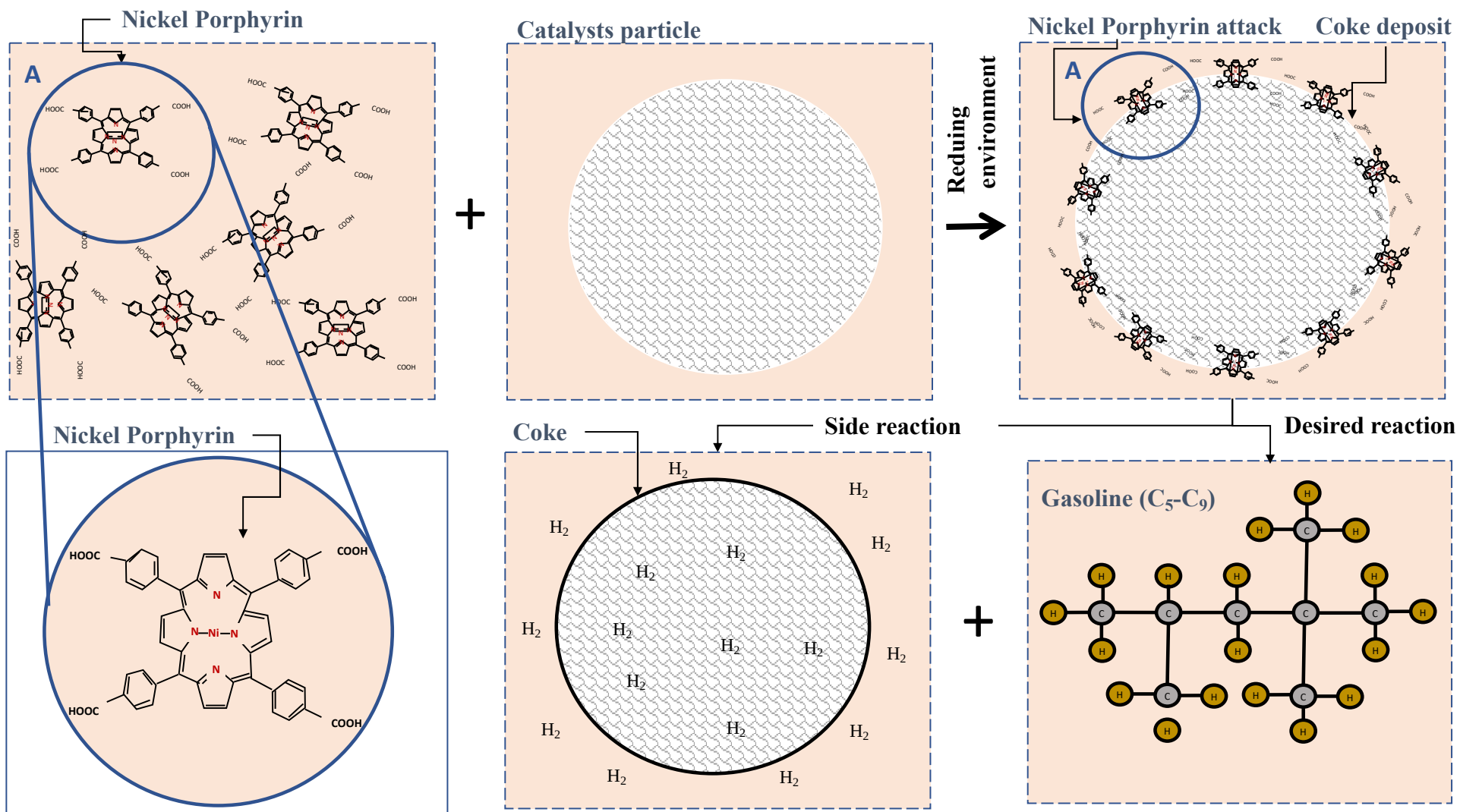
4.5 Nickel poisoning

The deactivation of RFCC catalysts by nickel occurs by catalysing unwanted side reactions and degrading any residual material in which it may be isolated. The nickel, acting as a catalyst for dehydrogenation, leads to more dehydrogenation activity than vanadium. However, nickel does not contribute to zeolite destruction [11,122]. Equation 1 presents the typical dehydrogenation reaction of cyclohexane to benzene in the presence of a nickel catalyst, as Al-Ayed and Kunzru reported [142].



The dehydrogenation, however, affects the performance of the RFCC unit due to the excessive formation of coke and hydrogen by shifting the selectivity from the desired cracked products. The coke derived from the catalysed dehydrogenation by contaminant metals such as nickel and iron is called contaminant coke.

The mechanism of nickel deactivation of RFCC catalysts is illustrated in Fig. 8. In the reactor during cracking, nickel in the form of porphyrin in the feed is first deposited on the catalyst surface, possibly in an amorphous state, deactivating it locally by covering the surface. Secondly, the surface layer of the nickel deposit gets thicker with time via the formation of crystalline sulfides migrating into the porous structure. The sulfides eventually destroy and deactivate the catalyst pores leading to undesired reactions such as dehydrogenation and coke formation.



707 Fig. 8 An illustration of the mechanisms of the nickel poisoning catalysts.

708 4.6 Vanadium poisoning

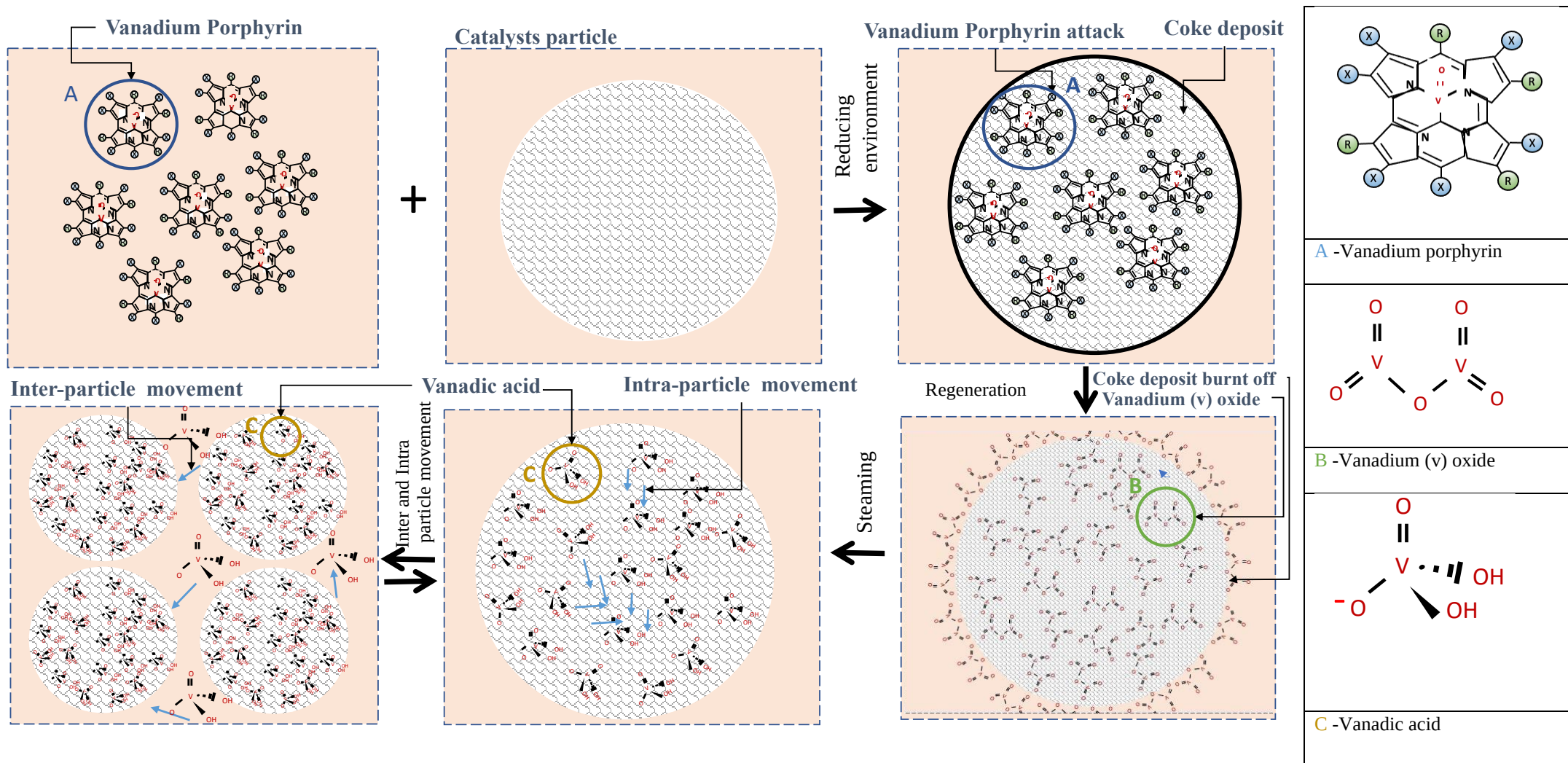
709 Numerous studies have proposed various means of vanadium attack on the RFCC
710 catalysts [15,16,120–122,143]. One of the routes was a study of several catalysts and
711 characterisation techniques such as BET, XRD, and electron microprobe [101,143]. The
712 mechanism of vanadium deactivation of RFCC catalysts is illustrated in Fig. 9. In the reactor,
713 during cracking, vanadium in the form of porphyrin in the feed is first deposited throughout
714 the catalyst particle and reduced to vanadium. Secondly, in the regenerator, it is oxidised to
715 vanadium pentoxide (Equation 3), which in the presence of steam, forms a highly mobile
716 vanadic acid (Equation 4). H_3VO_4 is a strong acid with a dissociation constant (pKa) of 0.05
717 [144].



718 This acid causes the mobility of vanadium and the subsequent attack on the zeolite
719 structure. The vanadic migrates into the zeolite structure and causes hydrolysis of the silicon-
720 aluminium bond leading to the collapse of the zeolite structure.

721 It is costly to design a catalyst regeneration system to prevent the formation of
722 vanadic acid. Hence, the effect of vanadium destruction on RFCC catalyst can be controlled
723 by developing appropriate additives to trap the vanadic acid. Appropriate traps are designed
724 based on the principle of acid-base chemistry, which allows for the immobilisation of the
725 active vanadium species [145].

726 In addition to destroying the catalyst structure, vanadium also promotes
727 dehydrogenation reactions like nickel, leading to the formation of a large amount of hydrogen
728 and coke [146]. Etim et al. [147] reported that the dehydrogenation effect of



729 Fig. 9 An illustration of the mechanisms of the vanadium poisoning catalysts.

730 Table 6: Mitigation Strategies of Metal Poisons

Feed Metal	Deposition Method	Primary Effect	Secondary Effect	Problems	Solutions	References
Iron	Peripheral deposition. Low mobility between particles	Creates nodules on the surface of the particle at very high levels	Dehydrogenation agent, CO promotion and Catalyst circulation problems	• Circulation issues and lower ABD • Over-promotion in partial burn units • Slightly increased SO_x emissions	• Flush catalyst • Catalyst change	[134,135]
Sodium	Deposits evenly and does not migrate	Permanent catalyst poison	Forms eutectic with vanadium	• Decreased catalyst activity	• Flush catalyst • Catalyst change • Optimise desalter operation	[140,141]
Nickel	Peripheral deposition. Does not migrate	Dehydrogenation agent (promotes coke and H ₂ production)	--	• Increased H₂ and coke production	• Antimony injection • Catalyst change • Flush catalyst	[122,148]
Vanadium	Deposits evenly and migrates in the regenerator	Permanent catalyst poison	Dehydrogenation agent	• Increased H₂ and coke production • Decreased catalyst activity	• V-trap additives such as Tin compounds • Flush catalyst	[101,145]

vanadium is one-fourth that of nickel when vanadium and nickel are deposited in equal amounts on the cracking catalyst. It was suggested by Tangstad et al. [149] that vanadium in +4 and +5 oxidation states have similar dehydrogenation activities, while vanadium in +3 or lower oxidation states has almost no dehydrogenating effects.

5 Mitigation Strategies of Metal Poisons

The mitigation strategies for metal (iron, sodium, nickel and vanadium) poisoning are presented in Table 6. As can be seen, the deposition methods, the primary and secondary effects, and the challenges associated with these metals vary. A detailed description of the metal poisons' mitigation processes is given in 5.1 to 5.4.

5.1 Iron mitigation

To mitigate the effect of iron, the refiner should employ a fresh catalyst with optimised surface porosity. A minimal catalyst surface pores will usually lead to poor diffusion of feed, while optimised catalyst surface pores have excellent feed diffusion. In minimal catalyst surface pores, the threshold for added iron is usually low in the range of 0.3 wt% compared to optimised catalyst surface pores, with a threshold of added iron of 1.5 wt% and above. These optimised catalyst pores make the catalyst withstand the harmful effect of iron by providing additional pores to accommodate the pore-blocking effect of iron. Another way to mitigate the detrimental impact of iron is to increase the fresh catalyst additions into the regenerator to flush iron out of the circulating inventory. This method is usually very expensive to practice due to the high cost of catalysts, as it consumes a lot of cracking catalysts [150,151]. Typically, catalyst consumption is 1200 tons/day, but using this method, the consumption could be as high as 2000 tons/day. Therefore, this approach must be

optimised with iron poisoning as a potential objective function and catalyst cost as a constraint [152].

5.2 Sodium mitigation

Sodium acts as a permanent catalyst poison; therefore, a fresh catalyst with low sodium should be employed to mitigate the effect of sodium. Another reliable way to reduce the sodium effect is to improve and optimise desalter operations to remove sodium from the feed. Refineries such as the Dangote petroleum refinery and petrochemicals in Nigeria employ a double desalter in their crude distillation unit to tackle the sodium effect. Sodium can also be mitigated by increasing the catalyst addition to flush sodium out of the circulating inventory. This increases fresh catalyst activity to combat activity loss from sodium. However, this method is not sustainable as it is very expensive to practice due to the high cost of catalysts [153].

5.3 Nickel mitigation

Ni becomes a concern at an equilibrium catalyst concentration greater than 800 ppm [154]. Some refiners have used antimony (Sb) injection to passivate the harmful effect of nickel. However, antimony has been found to increase NO_x emissions, bottom fouling and cause some environmental concerns regarding the safe handling of antimony [154].

A more reliable way to control nickel's effect is using speciality alumina, bismuth and rare earth compounds. Alumina and bismuth passivate nickel by forming nickel alloy on the catalyst surface, which is difficult to reduce to more detrimental elemental nickel. While these have proved successful to some extent, speciality alumina and bismuth are immobile under RFCC conditions. They cannot migrate to the nickel that deposits peripherally on the catalyst's surface, reducing their effectiveness. Rare earth compounds, on the other hand, are

very expensive for application in the refinery despite their success at mitigating the harmful effect of nickel.

The shortcomings of antimony, speciality alumina, bismuth and rare earth compounds are overcome using boron-based technology, identified as the most reliable technology to mitigate the effect of nickel on RFCC catalysts. Boron is mobile under RFCC conditions, migrating to the nickel and occupying the interstitial spaces forming Ni-B alloy. This Ni-B alloy is difficult to reduce to a detrimental form under riser conditions, thereby mitigating the dehydrogenating effect of nickel. Nickel does not increase NO_x emissions and is safe to handle and environmentally friendly [155].

5.4 Vanadium mitigation

Vanadium typically becomes a concern at an equilibrium catalyst concentration greater than 1500 ppm [155]. Refiners can use vanadium additives to mitigate the harmful effect of vanadium. These additives include rare earth oxides (REO), calcium and magnesium-based. Tin has also been identified as having the potential to mitigate vanadium poisons. It is important to note that these additives are added separately or blended into the catalyst formulation during the synthesis.

Low sodium fresh catalyst should be employed to mitigate the effect of vanadium because the harmful impact of vanadium is exaggerated in the presence of sodium. Another way to mitigate the effect of vanadium is to increase fresh catalyst additions into the regenerator or use purchased catalysts to flush out the vanadium in the circulating inventory. While this method is successful, it is not advisable to practice because it is very expensive [155].

5.5 Passivators for the simultaneous mitigation of nickel and vanadium poisons

The various passivators and mitigation strategies of the individual metal contaminants (sodium, iron, nickel and vanadium) that affect the RFCC catalyst have been discussed in 5.1 to 5.4. Some metal passivators can also be combined into a single catalyst to utilise their synergistic effects for mitigating multiple metal poisons. Despite the importance of using a single passivator to address the deleterious effect of nickel and vanadium, far too little attention has been paid to this area of research. Hoffer et al. [6] studied heavy metal passivators/traps for the FCC process by combining antimony (Sb) and iron (Fe). The authors reported that when Sb/Fe passivator was incorporated into an FCC catalyst contaminated with nickel, H₂ production was reduced by 15%. When the same combination was incorporated in the FCC catalyst deactivated with vanadium, H₂ production declined by 20%. Finally, when these metals combination was used to passivate FCC catalyst co-deactivated with nickel and vanadium, H₂ and coke production was reduced by 30%. The authors concluded that the combination of Sb and Fe successfully mitigated the deleterious effects of nickel and vanadium. However, Fe is a dehydrogenating poison and is unsuitable for mitigating the dehydrogenation effect of nickel and vanadium.

In a similar vein, Hayes and Castillo [156] used Sb/Sn to passivate nickel and vanadium-contaminated cracking catalysts. The findings suggested that H₂ yield decreased by 17% compared to 10% when Sb alone was used. It was concluded that a combination of Sb and Sn produces even better passivation than Sb when used alone. However, Sb as a passivator was found to increase NO_x emission and causes fouling of downstream equipment.

6 Metal Passivators (Traps) for the Inhibition of Nickel and Vanadium on RFCC Catalyst

Overall, amongst the metals and non-metal contaminants presented earlier in Table 5, nickel and vanadium are more harmful because when they are deposited on the catalyst, they interact with the catalyst to change the main reaction pathways and establish parallel reactions such as dehydrogenation, causing a shift in the catalyst selectivity (Fig. 8 and Fig. 9). These metals actions, therefore, become challenging in petroleum refining relying on the RFCC process and so, negatively impact the refining cost.

It should be noted that “passivator” and “trap” are not used interchangeably and that the mixture of metal oxides comprises components that either passivate or trap the metal impurities. “Passivator” is a structure that decreases the activity of undesirable metals such as nickel and vanadium. A “trap” is a composition that immobilises contaminant metals that have both intra-particle and inter-particle mobility in the RFCC catalyst mixture [6]. Such mobile metals include vanadium and sodium. A passivator may not necessarily immobilise vanadium, and a trap certainly may not passivate vanadium. The passivation process works by forming a thin film layer of oxides on the catalyst’s surface, which prevents the catalyst’s ability to produce contaminant H₂ and coke in the presence of contaminant metals during the RFCC process.

The process descriptions and key findings of various passivators of vanadium and nickel contaminants in the RFCC process are presented in Table 7. The main passivators of vanadium poisoning are tin compounds, rare earth compounds and alkaline earth compounds, whereas typical nickel poisons are antimony oxide, cerium oxide, bismuth, and boron. Detailed discussions on the effect of these passivators on nickel and vanadium are given in Table 7 and discussed in sections 6.1 and 6.2 below.

844 Table 7: Nickel and vanadium contaminants and their passivators/traps.

Contaminants	Passivators	Process conditions	Key findings	Ref
Nickel	Antimony Compounds	Materials: FCC catalyst; nickel Naphthenate; Antimony oxide Drying: Temp: 204°C, Time: 1 hr. Calcination: Temp-677°C, Time- 3 hrs. Steaming: Type - 100% steam, Temp - 732°C, Time - 8 hrs.	<u>Impregnation with Nickel</u> • Decrease in gasoline yield • Increase in H₂ and coke <u>Passivation with Antimony</u> • Decrease in surface nickel atoms • Reduction in coke and dry gas • Formation of Ni-Sb alloy	[154]
		Materials: REY-zeolite; nickel Naphthenate; Antimony oxide Nickel loading: 0.1 – 1.0 wt% Drying: Temp- 100°C, Time - 1 hr. Calcination: Temp - 590°C, Time - 14 hrs.	• Increase in H₂ and coke yield with increasing nickel loading up to 1% • The addition of antimony reinstates the transformation and production of H₂ and coke otherwise obtained with an unpoisoned catalyst • A strong decrease in percentage conversion on the addition of nickel to the catalyst	[7]
		Materials: FCC catalyst; Nickel naphthenate; vanadium naphthenate; antimony compound Steaming: Type – 100% steam, Temp – 766°C, Time -18 hrs. Calcination: Temp – 677°C, Time – 2 hrs.	• 30% loss of catalyst surface area after steaming • All catalysts with metal traps showed a boost in gas oil conversion • The presence of a metal trap increases the research octane number (RON) of the gasoline • Decrease in H₂ and coke for catalysts with metal traps	[157]
	Rare Earth Compounds	Materials: FCC catalyst; nickel naphthenate; rare earth oxides Drying: Temp: 80°C, Time - 1 hr. Calcination: Temp: 600°C, Time - 2 hrs. Steaming: Type - 100% steam, Temp - 800°C, Time: 10 hrs.	• Nickel alone does not affect zeolite surface area loss • 3000 ppm of vanadium decreases the zeolite surface area • There was 59% preservation in zeolite surface area with the introduction of rare earth oxides • 44% surface area preservation by rare earth oxides when the catalyst was poisoned by both vanadium and nickel • The presence of nickel reduces the effectiveness of rare earth oxides.	[120]
		Materials: FCC catalyst; nickel compound; cerium octoate Equilibration: Temp: 649°C, Time: 0.5 hr. Calcination: Temp: 649°C, Time: 1 hr. Steaming: Temp - 788°C, Time: 6.5 hrs.	• Nickel poisoning causes a high yield of H₂ gas • The addition of cerium to the catalyst greatly decreases the amount of hydrogen produced • The catalysts passivated with cerium resulted in lower coke values when steamed at 732°C or 788°C	[158]
	Bismuth	Materials: Nickel compound, vanadium compound, and bismuth compound Riser: Temp – 593°C, Pressure: 3.50 kg/cm ² Contact Time: 15 sec. Catalyst-to-oil ratio: 4:1 Regenerator: Temp – 682°C, Time: 3 – 30 mins.	• Decrease in conversion and gasoline yield with nickel and vanadium poisons • Increase in H₂ and coke yield with nickel and vanadium poisons • Decrease in yield of H₂ and coke upon introduction of bismuth • Conversion and gasoline yield increase with the addition of bismuth	[159]
		Materials: FCC catalyst; nickel naphthenate; bismuth compound; antimony compound Drying: Temp: 120°C, Time - 1 hr. Calcination: Temp - 550°C, Time: 5 hrs.	• Bismuth and antimony suppress the harmful effect of nickel • Replacement of antimony with bismuth eliminates hazardous material.	[160]

Vanadium	Boron	Materials: USY-Zeolite; nickel octoate; boron oxide Drying: Temp - 100°C, Time: 1 hr. Calcination: Temp - 650°C, Time: 3 hrs.	- Characterisation of boron is difficult because of its low atomic weight. [161] - CO DRIFT with three pretreatment temperatures can characterise the impact of boron on the catalyst - Boron-based technology passivates nickel by decreasing its reducibility under typical FCC conditions at less than 600°C.
		Materials: FCC catalyst; nickel naphthenate; vanadium naphthenate; boron compound Drying Temp - 80°C, Time - 2 hrs. Calcination: Temp - 550°C, Time - 5 hrs. Steaming Temp: 760°C, Time: 6 hrs.	- Operating H₂/CH₄ decreased with boron at constant equivalent Ni [155] - Increase in operating gasoline yield with boron at constant conversion 3% LPG and gasoline yields, higher conversion with boron 26% reduction of unwanted hydrogen with boron 16% coke reduction with boron
	Tin compounds	Materials: Zeolite-Y; vanadyl naphthenate; tetraphenyl Tin. Drying Temp.: 100°C Calcination Temp: 540°C Steam: 95% Steaming Temp.: 730°C	<u>Impregnation with vanadium</u> [162] - Spectroscopic studies showed mobile V species on the zeolite surface - V₂O₅ was formed in large amounts after hydrothermal treatment - Irreversible loss of crystallinity leading to the collapse of the zeolite framework <u>Passivation with Lanthanum</u> - Retention of crystallinity on the addition of tetraphenyl tin after steam-ageing - Tin limits vanadium-zeolite interactions
		Materials: NH ₄ -Zeolite-Y; vanadyl naphthenate; tetraphenyl Tin. Drying Temp: 100°C Calcination Temp.: 575°C Calcination Time: 12 hrs. Steam: 95% Steaming Temp.: 730°C Steaming Time: 10 hrs.	- Vanadium deposited on the catalyst does not migrate initially until after steaming. [163] - The order of deposition of vanadium and tin on the zeolite catalyst is irrelevant - The presence of tin removes the zeolite-vanadium interaction by forming a tin-vanadium complex - Excess tin is to be avoided when using it to passivate vanadium
Basic alkaline earth compounds		Materials: NH ₄ -Zeolite-Y; vanadyl naphthenate; tetraphenyl Tin. Drying Temp.: 100°C Calcination Temp.: 575°C Calcination Time: 12 hrs. Steam/N ₂ : 95% steam & 5% N ₂ Steaming Temp.: 730°C Steaming Time: 10 hrs. Flow rate of water: 4ml/hr.	- Vanadium present as free VO²⁺ cations favours the breakdown of zeolite lattice [164] - The order of deposition of tin and vanadium has little effect on the nature of the resulting species
		Materials: FCC catalyst; vanadyl naphthenate; Calcium Chloride. Drying Temp.: 110°C Drying Time: 3 hrs. Calcination Temp.: 540°C Calcination Time: 2 hrs. Steam/Argon: 4% steam & 96% Argon Steaming Temp.: 760°C Steaming Time: 5 hrs.	- The addition of vanadium to the FCC catalyst destroys the structure of the zeolite and reduces its activity [165] - The addition of calcium decreases the yield of coke and gas with increasing gasoline selectivity

	Materials: zeolite catalyst; vanadyl naphthenate; magnesium acetate Drying Temp.: 200°C Drying Time: 1 hr. Calcination Temp.: 720°C Calcination Time: 1 hr. Steam/Air: 60% steam & 40% air Steaming Temp.: 760°C Steaming Time: 2 hrs.	• Increase in coke and gas factor after impregnation of the catalyst with vanadium • The introduction of Mg leads to a reduction in coke and an increase in gasoline • The introduction of Mg leads to the preservation of the zeolite crystal structure.	[143]
Rare Earth Compounds	Materials: FCC catalyst; vanadium naphthenate; Lanthanum octoate Equilibration Temp.: 770°C Equilibration Time: 15 hrs. Steam: 100% Impregnation Temp.: 60°C Calcination Temp.: 600°C Heating rate: 2°C/min. Calcination Time: 5 hrs. Materials: USY-Zeolite; ammonium vanadate; lanthanum oxide. Drying Temp.: 80°C Drying time: 24 hrs. Calcination Temp.: 500°C Calcination Time: 2 hrs. Steaming Temp.: 800°C Steaming Time: 4 hrs.	<u>Impregnation with vanadium</u> • Shrink in zeolite unit cell size from 24.58 to 24.26 Angstroms due to dealumination • vanadium is mostly concentrated on zeolite, which is transformed into mullite and cristobalite after hydrothermal treatments. <u>Passivation with Lanthanum</u> • Zeolite crystal was kept higher than 90% Zeolite and the entire matrix were preserved from the effect of vanadium <u>Impregnation with vanadium</u> • Higher vanadium deposition (1.12 wt%) induced intense damage to the zeolite crystal structure. • Catalyst acidity decreases with an increase in vanadium content. • The residue conversion decreased sharply between 86.7 wt% and 70.4 wt% with a change in the concentration of vanadium contaminant from 0 ppm to 5000 ppm. • The yield of target products (LPG + cracked gasoline) declined by 18.68 wt%, and the light cycle oil (LCO) increased significantly. <u>Passivation with Lanthanum</u> • The by-product yield, including coke, dry gas, and cycle oils, nearly stayed the same, besides cracked gasoline and LPG yields • Adding cerium oxide and lanthanum oxide increased the catalysts' surface area and feed conversion. • The introduction of lanthanum oxide into the catalyst mix impeded the destruction by highly deposited vanadium	[166]
			[167]

6.1 Passivators for nickel poisoning

6.1.1 Antimony (Sb) as nickel passivator

The use of antimony compounds as nickel passivator has been studied by several authors, as detailed in Table 7 [7,154,157]. In general, the source poison is nickel in the form of naphthenates, while antimony oxide is employed as the passivating compound. During the impregnation process of antimony compounds on the FCC catalyst, the drying temperature has been established to range between 100 – 204°C for 1 hour, calcination temperature in the range of 590 – 677°C for 2 – 14 hours and all catalysts are steam treated in 100% steam environment at a temperature ranging between 732 – 766°C for 8 – 18 hours.

Corma et al. [154] studied nickel passivation on fluidised cracking catalysts with different antimony complexes by impregnating the FCC catalyst with nickel naphthenates and impregnating the same with antimony oxide as passivator. The authors reported that when FCC catalyst was impregnated with nickel naphthenates, gasoline yield decreased by 5 wt% while H₂ and coke increased by 0.5 wt%. Upon passivation of the nickel-deactivated catalyst with antimony oxide, the formation of Ni-Sb alloy led to a decrease in coke and dry gas production of 16% and 8%, respectively.

Goldwasser et al. [7] studied Ni-Sb interaction in a rare earth Y zeolite by impregnating FCC catalyst with nickel naphthenate and passivating the nickel-deactivated catalyst with antimony oxide. It was reported that there was an increase in H₂ and coke yield with increasing nickel loading of 1%. A strong decrease in percentage conversion was also observed with the addition of nickel to the catalyst. The addition of an antimony compound reinstated the transformation and production of H₂ and coke otherwise obtained with an active unpoisoned catalyst.

Antimony, when used as a nickel-passivator, demonstrated a substantial decrease in the nickel poisoning of an RFCC/FCC catalyst [16]. The interaction of antimony with nickel has been studied extensively in the past four decades. FCC catalysts with a nickel loading range of 1.9 and 4.4% weight and Sb to Ni ratios between 0.00 and 0.41 were examined [168]. The X-ray diffraction findings suggested that Ni-Sb solid matrices are formed with an increased level of Sb appearing on the nickel surface. In explaining the results, however, geometric and electronic effects were reported as follows:

- (i) Alteration of nickel surface atoms: the electronic properties of nickel surface atoms were altered by antimony, so their catalytic activity decreased significantly.
- (ii) Quantity of antimony available for passivation: the equilibrium between the interaction of antimony, nickel and support determines the amount of antimony available to passivate nickel.
- (iii) Blockage of nickel sites: antimony geometrically blocks nickel sites in the catalyst.

The interaction between nickel and antimony was studied in a Rare-Earth (Lanthanum oxide)-exchanged zeolite-Y [7]. Furthermore, the influence of the addition of antimony on the structure of nickel and chemisorptive properties and structure of nickel was investigated through temperature programmed reduction (TPR) studies. It was found that the addition of antimony intensely lowered the chemisorption of H_2 due to the formation of nickel-antimony alloy, which is difficult to reduce to more detrimental elemental nickel under the riser condition of RFCC operation.

It was found by Keeley et al. [169] that apart from passivating nickel, antimony will also poison carbon monoxide (CO) combustion promoters and can increase nitrogen oxide (NO_x) emissions. There are also environmental and safe handling concerns for antimony. For

these reasons, antimony is not permitted in some countries. In addition, there is also the concern of slurry circuit fouling due to antimony injection.

6.1.2 Rare earth compounds as nickel passivator

Rare-earth compounds have been employed to passivate the deleterious effect of nickel [120,134]. Nickel naphthenates are the major sources of nickel poison, and rare-earth oxides such as cerium oxide and lanthanum oxide are the passivators used to mitigate the harmful effect of nickel. The operating conditions are similar to those used in antimony mitigation of nickel poisons. Drying temperature ranges from 80 – 100°C for 1 hour, calcination temperature ranges from 600 – 649°C for 1 – 2 hours, while steam treatment is carried out in a 100% steam environment at a temperature between 788 – 800°C for 6.5 – 10 hours.

Escobar et al. [120] investigated the interaction between Ni and V with ultra-stable-Y and rare-earth-Y zeolite during hydrothermal deactivation by impregnating FCC catalyst with nickel naphthenate and passivating same with rare-earth oxides. It was observed that nickel alone does not affect the loss in zeolite surface area except in the presence of vanadium. There was 59% preservation in zeolite surface area with the introduction of rare earth oxides and 44% surface area preservation by rare earth oxides when the catalyst was poisoned by both vanadium and nickel. It was concluded that the presence of nickel reduces the effectiveness of rare earth oxides.

An investigation into the passivation of FCC catalysts showed that when 3000 ppm of cerium oxide was used to passivate FCC catalysts contaminated with 3000 ppm of vanadium and 1500 ppm of nickel, hydrogen production increased by 2% instead of an expected decrease. In contrast, when 3000 ppm of cerium octoate was used to passivate the FCC catalyst, the production decreased by 6%. It was concluded that cerium in organic form (octoate) significantly reduced the amount of hydrogen produced, which is attributed to

nickel contamination [158]. However, cerium is highly expensive and hence not cost-effective. This limits its potential use.

6.1.3 Bismuth as nickel passivator

Due to the environmental challenges associated with antimony and the high cost of rare earth compounds as nickel passivators, bismuth was used as a replacement for both antimony and rare earth compounds to passivate the harmful effect of nickel. During the deactivation and passivation process, the drying and calcination temperatures are kept at 120°C for 1 hour and 550°C for 5 hours, respectively. The regenerator temperature was 682°C for 3 – 30 minutes, a catalyst-to-oil ratio of 4:1 and riser temperature, pressure and contact time were kept at 593°C, 3.50 kg/cm² and 15 seconds, respectively.

Bismuth compounds developed by Gulf petroleum have been reported as effective nickel passivating agents [103]. To the best of our knowledge, there is no published information on the chemistry of the interaction of bismuth with nickel. However, various reports have been published on the commercial benefits of bismuth additives compared to antimony [103]. Bismuth has been chosen as a preferred nickel passivating agent because it is less toxic than antimony, as the United States Environmental Protection Agency does not list it as a hazardous chemical.

Ramamoorthy and Krishna. [159] in their patent studied the method of suppressing metal contaminants' harmful effects on cracking catalysts in FCC by contacting a nickel-contaminated hydrocarbon feedstock with a cracking catalyst associated with a bismuth-containing metal passivating agent. The findings suggest that the suppression of nickel poisoning can be accomplished with a proportionally smaller amount of bismuth. In contrast, higher amounts of bismuth are less effective and provide adverse effects such as poor deposition efficiencies on the catalyst. Meanwhile, high bismuth levels can end up in cracked cycle oil products. These, in turn, can cause adverse effects on the downstream

hydroprocessing catalysts when the cycle oils are processed further. It was deduced that the best weight ratio of bismuth to nickel equivalents should be in the range of 0.01:1 to 1:1 in the feed to be effective [159].

6.1.4 Boron-based technology as nickel passivator

The deposition profile of nickel follows a particular gradient: nickel deposits peripherally on the catalyst's surface. Therefore, it requires a mobile passivator under RFCC conditions to migrate to the catalyst's surface and effectively passivate nickel. In addition to being environmentally friendly, boron is mobile under RFCC conditions, migrating to the nickel and occupying the interstitial positions in Ni-B alloy [169]. This alloy is difficult to reduce to a more detrimental nickel in the RFCC riser. Therefore, it passivates the harmful effect of nickel. Similar to the previous process conditions of other nickel passivators, the drying temperature is kept at 80 – 100°C for 1 – 2 hours. Calcination temperature and time of 540 – 550°C and 5 hours, respectively, while the steam treatment is done at 760°C for 6 hours in a 100% steam environment.

Boron-based catalysts can overcome the (i) NO_x emissions of antimony, (ii) high cost of cerium and (iii) immobility of bismuth. Boron is mobile under RFCC conditions, which makes it more efficient as it migrates within the catalyst to passivate nickel. BASF, a well-known catalyst manufacturing company, has studied boron chemistry to evaluate the Ni passivation mechanism. Based on multiple spectroscopic studies using boron-based technology, Ni can be kept in a less reducible state, which inhibits dehydrogenation reactions. Thus, it lowers the formation of hydrogen and contaminant coke. A boron-based technology as an innovative solution for RFCC performance improvement was studied [155].

A catalyst (BoroCat) was designed, and upon application of the catalyst to the RFCC unit, there was a 26% reduction in unwanted hydrogen, 16% coke reduction and a 3% increase in LPG and gasoline yields at higher conversion. A new spectroscopic methodology

was used in characterising the impact of boron interaction on RFCC catalysts contaminated with nickel via trend analysis of carbon monoxide diffuse reflective infrared Fourier transform spectroscopy (CO DRIFTS) [161]. The findings suggest that the characterisation of boron is difficult due to its low atomic weight. CO DRIFTS with three pre-treatment temperatures can be used to characterise the impact of boron on the catalyst. It was concluded that boron-based technology passivates nickel by decreasing its reducibility under typical RFCC conditions at a temperature of less than 600°C.

6.2 Passivators for vanadium poisoning

6.2.1 Tin compounds as vanadium passivator

Tin compounds have been identified as the most common passivators of the harmful effects of vanadium [162-164]. The source of vanadium poison and passivator has been established to be vanadyl naphthenate and tetraphenyl tin, respectively. The operating conditions for vanadium deactivation and passivation of FCC catalyst are within the same range as those of nickel, except that most of the steam treatments were carried out at 95% steam, with the remaining percentage being either N₂ or Argon.

Anderson et al. [164] in their study of tin passivation of vanadium in a metal-contaminated fluid cracking catalyst, carried out steaming in a 95% steam environment and 5% N₂ instead of 100% steam. The 5% inert medium makes the reactor system partially inert to prevent side reactions that will interfere with the steaming process. It was found that vanadium present as free VO²⁺ cations favour the breakdown of zeolite lattice. The order of deposition of tin and vanadium was found to exhibit a little effect on the nature of the resulting species that passivates the deleterious effect of vanadium.

Bradley et al. [162] carried out a spectroscopic study of tin passivation of vanadium contaminants in a model fluid cracking catalyst. It was observed that an irreversible loss of

crystallinity led to the collapse of the zeolite framework. Retention of crystallinity was detected after adding tetraphenyl tin after steam-ageing, and that tin was found to limit vanadium-zeolite interactions.

The passivation of vanadium using tin in metal-contaminated fluid cracking catalysts was conducted using tin Mossbauer spectroscopy. The work investigated the role of tin in the passivation of vanadium contaminants in model systems comprising aluminosilicate gel, EuY and EuY-gel mixtures [163]. Tin was added to the catalyst as tetraphenyl tin ($\text{Sn}(\text{Ph})_4$) and remained predominantly intact on the zeolite. However, it formed oxygen ligands on the gel replacing one or two phenyl groups. On the zeolite, a portion of the tin is left over on the surface while some portion migrates to internal sites. In the zeolite-gel mixture, tin is favourably adsorbed by the zeolite. It was observed that Sn (II), metallic tin, and vanadium-tin alloys were not produced at all stages of catalyst pre-treatment. Nonetheless, calcination and steaming formed vanadium/tin oxide complexes built over oxygen bridges in which tin is in the +4-oxidation state. The formation of $\text{Sn}^{4+}\text{-O-V}^{5+}$ linkages is possibly liable for the passivation of V by tetraphenyl tin deposition.

6.2.2 Rare-earth compounds as vanadium passivator

In addition to being able to passivate nickel, rare earth compounds such as lanthanum octoate, dysprosium and samarium naphthenates have proven successful at passivating the deleterious effect of vanadium [166,167]. The high cost of these metals may limit their potential use. The operating conditions for deactivation and passivation of FCC catalyst by vanadium and rare earth compounds are similar, as depicted in Table 7. The drying temperature is maintained at 80 – 100°C for 24 hours, calcination temperature of 500 – 600°C for 2 – 5 hours and finally, steam treatment is carried out in 100% steam for 4 hours.

A study on vanadium passivation with rare-earth compounds soluble in feedstock by impregnating FCC catalyst with rare-earth compounds such as lanthanum octoate,

dysprosium and samarium naphthenates at the same time with vanadium naphthenates was conducted [166]. Adsorption measurements exhibited that more than 90% of the zeolite structure is preserved after high-temperature hydrothermal treatment when $\text{La/V} = 1$. It was found that high-resolution analytical microscopy showed that rare-earth vanadates (LaVO_4 and SmVO_4) in the form of small particles are made on the surface of the catalyst grains on the catalyst matrix. These rare-earth vanadates were found both on the external surface and in the mesopores of the zeolite. The micro activity test indicated that gasoline yield was high, and coke and hydrogen yields remained higher than those of unpoisoned catalysts. The high gasoline level was due to the presence of rare-earth compounds to passivate the destructive effect of vanadium, while the high amounts of hydrogen and coke were due to the presence of vanadium which causes dehydrogenation and coke formation.

6.2.3 Basic alkaline earth compounds as vanadium passivator

Basic alkaline earth compounds such as calcium chloride and magnesium acetate have been investigated and found effective at scavenging vanadium [143,165]. The operating conditions are similar to those previously discussed for vanadium passivation using rare-earth compounds in 6.2.2, except that, in this case, the steam treatment was carried out in a 4% steam environment and 96% Argon. A small percentage of steam is employed, possibly because the cracking catalyst has undergone a certain level of deactivation before being used for this purpose. Therefore, it did not require a high percentage of steam. The Argon was used to provide an inert environment for the deactivation process.

An investigation was conducted into the mechanism and poisoning of vanadium-tolerant catalysts [143]. The findings show that basic alkaline earth oxides such as MgO and CaO are effective for vanadium scavenging. The micro-activity testing exhibited excellent activity retention when 20% MgO is blended with a cracking catalyst at vanadium loadings of 0.67% and 1.34% by volume on the catalyst. It was, however, concluded that the SO_x in

the regenerator flue gas can form a sulphate competing with vanadate formation. The degree of the competition is thermodynamically controlled since vanadate formation from the oxide expands the lattice and pores. This is similar to the observation in the case of the reaction of calcium oxide with sulfur oxides.

Chao et al. [165] investigated vanadium passivation using calcium by imaging secondary ion mass spectroscopy. Fresh FCC catalyst was impregnated with 2300 ppm of nickel and 4500 ppm of vanadium using a cyclohexane solution of nickel and vanadyl naphthenates. Calcium and clay were introduced into the Ni and V-loaded catalyst and tested. The performance test showed that adding Ca/clay to the catalyst decreases the yields of coke and gas with increasing gasoline selectivity. The findings also indicated that calcium, an alkaline-earth metal, can passivate the deleterious effect of vanadium. This may be due to the ability of alkaline earth metals to form a stable complex with vanadium and helps prevent the deleterious effect of vanadium.

7 Challenges and Prospects

7.1 Challenges and future research and development

Residue fluid catalytic cracking (RFCC) technology is the key to processing crude oil and unconventional feedstocks due to its ability to handle heavy crudes and maximise gasoline and propylene production. However, processing heavy and dirty crudes come with some challenges due to heavy metals, predominantly nickel, vanadium, sodium and iron. These heavy metals contribute to the following challenges:

- (i) Destruction of catalyst and the promotion of competing and undesirable side reactions: leads to the production of a large amount of hydrogen and coke.
- (ii) Production of a large amount of hydrogen: puts too much stress on the wet gas compressor in the RFCC unit, which is volume limited.

1065 (iii) Production of a large amount of coke: this upsets the reactor and regenerator heat
1066 balance and affects the main air blower in addition to blocking the catalyst's active
1067 sites, thereby reducing the activity of the catalyst.

1068 Based on our analysis, future research and development should focus on the
1069 following:

1070 (i) *Design of RFCC catalyst*: RFCC catalysts that can withstand the action of metal
1071 poisons present in heavy crude oils should be properly designed. Vanadium destroys
1072 the framework of the conventional FCC catalyst by hydrolysis of the silicon and
1073 aluminium bonds in the zeolite structure. Nickel promotes undesirable side reactions,
1074 and sodium neutralises the acid sites of the catalyst, which is responsible for cracking
1075 the hydrocarbon feedstocks.

1076 (ii) *Replacement of antimony as nickel passivator*: Antimony has been employed to
1077 mitigate the harmful effects of nickel poison due to its ability to form Ni-Sb alloy,
1078 which is difficult to reduce to a more deleterious elemental nickel. However,
1079 antimony has recently been identified to cause NO_x emissions. This constrained it due
1080 to stringent environmental regulations. In addition to antimony being responsible for
1081 the corrosion of downstream equipment due to its excessive injection, there are also
1082 safe handling concerns of antimony due to its toxicity.

1083 (iii) *Developing an RFCC catalyst that incorporates boron as a nickel passivator*: Unlike
1084 antimony, boron is mobile under RFCC conditions and can migrate to the immobile
1085 nickel on the catalyst, effectively passivating the harmful effect of nickel. Also, boron
1086 does not cause NO_x emissions and is environmentally friendly and safe to handle.
1087 These positive attributes of boron make it suitable as a nickel passivator and can
1088 therefore be incorporated into the catalyst mix during the synthesis process.

(iv) *More research into the passivation of vanadium.* Vanadium is regarded as the most harmful of all catalyst poisons. In addition to promoting dehydrogenation reactions, vanadium causes hydrolysis of the silicon and aluminium bond of the zeolite catalyst, which leads to the collapse of the catalyst structure. To date, tin-based compounds have been identified as having the ability to mitigate the harmful effect of vanadium, but there has not been much research in this area. Therefore, more research focus is needed in the area.

8 Conclusions

The Fluid Catalytic Cracking (FCC) process has been an area of interest to petroleum refiners due to its ability to convert the bottom of the barrel into more useful products leading to an increase in refinery margins. The change in crude oil quality towards heavier and unconventional crudes has necessitated the invention of Residue Fluid Catalytic Cracking (RFCC) technology along with its associated catalysts, which can withstand the high coke precursor (Conradson Carbon Residue) and heavy metals such as vanadium and nickel that come with the heavy and dirty crudes. Conradson Carbon Residue (CCR) generates a large amount of coke, leading to high temperature in the regenerator when this coke is burnt off the catalyst during the regeneration process. Nickel and vanadium promote undesirable side reactions such as dehydrogenation and destruction of the cracking catalyst by hydrolysis of the silica-alumina bond, respectively. This review summarises nickel and vanadium mitigation strategies, the most important metals that affect the cracking catalyst. The advantages and drawbacks of nickel and vanadium mitigation strategies were evaluated. Antimony has been extensively reported to be successful at mitigating the harmful effect of nickel. Still, it causes NO_x emission and corrosion of equipment due to excessive injection. There are also safe handling concerns. Bismuth and transition alumina have been used as replacements for antimony, but bismuth is not very effective, and transition alumina is not

mobile under RFCC conditions. The mitigation of vanadium has not been widely studied. However, tin-based compounds have been proven to successfully mitigate vanadium's effect. As a result of the existing challenges of nickel mitigation, as discussed in this review, boron has been identified as an environmentally benign and economically effective mitigation strategy for a nickel. Boron does not cause NO_x emission. It is safe to handle and mobile under RFCC conditions, making it a promising alternative for mitigating nickel poisoning.

Credit author statement

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